



Delta State University Abraka

DELSU

POST-UTME

PAST QUESTIONS

AND

ANSWERS

(2018 EDITION)

JAMBITES OF NIGERIA

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DELSU POST-UME PAST QUESTIONS AND ANSWERS (2018 EDITION)

PREFACE

Jambites Of Nigeria strives to continue her vision of providing assistance and up-to dates of information about all schools to every Nigerian student, especially admission seekers and high school Leavers.

In our quest to remain, maintain and sustain our current status as the best and ever first corporate organization that manage the entirety of Nigerian undergraduates, we had in the past one year embarked on massive compilation, revision, analysis and research on previous POST-UTME examinations conducted, the feasibility studies and chances of a candidate passing or and of course failing the test.

From our research findings, we have found without further verifiable hypothesis that the POST-UTME Past questions still remain a secret to passing through these tests in success.

We therefore, present and recommend these past questions for all candidates preparing for this examination. Our idea in this sense does not focus on duplicating the efforts made by several other Nigerians, Lecturers, teachers and tutors in this direction. We only aimed at correcting some of their shortfalls, and most importantly; making these study, revision and preparatory materials readily available to everyone in any part of the country.

The questions contained herein has been drawn from the candidate's school of choice in the relevant courses to facilitate easy and more result orientated performance. In this respect, actual and real examination questions were extracted from your school of choice and is therefore being presented to you, Please read and study through to succeed.

We wish all Nigerian students goodluck in their pursuit of academic excellence.

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NOTE: Copies of **DELSU** and other **Schools Post-Utme Exam Past Questions and Answers** are available for download on our website. Recommend us to your friends today by giving them our [website](http://www.jambitesofnigeria.com).

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Regards!

**DELTA STATE UNIVERSITY, POST-UTME SCREENING PAST
QUESTIONS AND SOLUTIONS**

**ANSWER ALL QUESTIONS RELEVANT TO YOU:
SHADE THE ANSWER SHEET AS APPROPRIATE WITH HB PENCIL
ONLY**

ECONOMICS 2007/2008

1. For two substitute goods, the gross elasticity of demand is
 - A. greater than one but less than
 - B. zero
 - C. negative
 - D. positive
 - E. infinity

2. If a person receives a higher wage than would be necessary to induce him to work, he is said to be receiving
 - A. rent
 - B. profit
 - C. interest
 - D. gain
 - E. moratorium

3. Which of this is likely to be inflationary
 - A. tax increase
 - B. budget surplus
 - C. increase in unemployment
 - D. wage increase
 - E. increase in labour supply

4. A characteristic of a debenture is that
 - A. its yield is based on profits
 - B. its yield is fixed rate of interest
 - C. it has no redemption date
 - D. there is a voting when interest is paid
 - E. its interest is variable

5. Which of these is not include in measuring national income by the income approach
 - A. wages and salaries of public servants

- B. students bursaries and scholarship
 - C. profit of international
 - D. income of self employed person
 - E. rents on property
6. The revenue accruing to the seller of commodity X as a result of a fall in price will
- A. increase
 - B. fall
 - C. first fall but increase later
 - D. remain unchanged
 - E. change noticeably
7. Deflation is persistence fall in price in the general level and is usually caused by
- A. a reduction in total demand
 - B. an increase in government spending
 - C. an increase in money supply to banks
 - D. an increase in aggregate demand
 - E. an increase in money supply relative to demand
8. Under partnership, investors who have no desire to be actively involved on the day-to-day management of such organizations are called
- A. stock brokers
 - B. sleeping partners
 - C. part-time investors
 - D. ordinary partners
 - E. debenture holders
9. In a sole proprietorship, decisions are made by the
- A. government
 - B. board of directors
 - C. management
 - D. shareholders
 - E. owners
10. Which of the following is regarded as fixed cost?
- A. Cost of raw materials
 - B. Labour wages

- C. Rent on Land
 - D. cost of light
11. Demand as a factor of production is
- A. a composite demand
 - B. a joint demand
 - C. a derived demand
 - D. an elasticity of demand
 - E. cross elasticity of demand
12. One disadvantage of sole proprietorship is its
- A. limited liability
 - B. high profits
 - C. high sense of ownership
 - D. low credit rating
 - E. low failure value
13. The monetary system that requires double coincidence of wants is known as
- A. the gold standard
 - B. the uarter practice
 - C. the commodity system
 - D. the good exchange standard
 - E. the Cheque system
14. The marginal theory of distribution makes an assertion that the price of any factor depends upon its marginal
- A. utility
 - B. productivity
 - C. rate of substitution
 - D. revenue
 - E. proceeds
15. By using exchange controls, a country tries to eliminate a balance of payments deficit by
- A. limited her imports to its currency of exports
 - B. reducing the nation's domestic price level
 - C. limiting her exports to its currency value of imports
 - D. overvaluing the country's

- E. undervaluing the country's currency
- 16. Economies of scale operate only when
 - A. marginal cost is falling with input
 - B. average cost is falling with output fixed cost is variable
 - C. fixed cost in variable
 - D. variable cost less than fixed cost
 - E. variable cost is equal to cost of output
- 17. If an increase in income induces a reduction in the demand for beans, beans can be referred to as
 - A. a normal good
 - B. an inferior good
 - C. a substitute
 - D. a dash
 - E. a gifted good
- 18. Optimum population is desired because it enable an economy to attain Maximum
 - A. per capital income
 - B. per capital output
 - C. per capital real income
 - D. per capital revenue
 - E. growth rate
- 19. Marginal cost curve intersects average cost curve
 - A. from above at its lowest point
 - B. from below before the lowest point
 - C. from below at its lowest point
 - D. from below after the lowest point
 - E. at the zenith of turn
- 20. The concept of economics efficiency refers to
 - A. obtained the maximum output from available resources at the lowest possible cost
 - B. conservation of natural gas and oil deposits
 - C. equity in the distribution of national resources
 - D. production without waste
 - E. the limited wants unlimited resources dilemma

21. A major factor affecting the value of money is the
- A. price level
 - B. banking habit
 - C. transaction motive
 - D. divisible nature of money
 - E. transferability
22. Which of the following is an example of invisible item on balance of payment account?
- A. shipping and aviation
 - B. import and export
 - C. merchandise
 - D. bullions
 - E. tariffs
23. The best index for international comparison of standard of living is the
- A. cross national product
 - B. net national income
 - C. per capita income
 - D. cross national income
24. A major obstacle to economic development in many African countries is
- A. a rise in industrial output
 - B. low farm productivity
 - C. free trade
 - D. free interaction
 - E. controlled international relations
25. The most popular adopted industrialization strategy in West Africa is
- A. import substitution
 - B. export promotion
 - C. ports development
 - D. infant industries protection.

ECONOMICS 2007/2008 SOLUTIONS/ANSWERS

- | | | | | |
|-----|-----|------|------|------|
| 1.D | 6.C | 11.C | 16.C | 21.A |
| 2.E | 7.A | 12.D | 17.B | 22.A |

3.D	8.B	13.B	18.B	23.C
4.B	9.E	14.B	19.C	24.B
5.B	10.C	15.A	20.A	25.B

MATHEMATICS 2007/2008

1. Simplify 12^6

- A. $2 \cdot 6$
- B. 2
- C. 6
- D. $12 \cdot 6$
- E. $6 \cdot 6$
- 24 - 6

2. Without using tables, the numerical value of $\log_7 49 + \log_7 (1/7)$ is

- A. 0
- B. -1
- C. 1
- D. $\frac{1}{2}$
- E. 36

3. If $x^2 - 6x + 1 = 0$, has coincident roots, the value of n is

- A. 9
- B. -9
- C. 3
- D. 4
- E. 36

4. The roots of $y^2 - y - 12 = 0$ are

- A. 4, -3
- B. -4, 3
- C. -4, -3
- D. 4, 3
- E. 36

5. $200 - 12$ is

- A. 22
- B. 72

- C. -22
- D. 2
- E. 42

6. If $8x/2 = 23/8 \times 43/4$ the value of x is

- A. 1
- B. $11/4$
- C. $15/8$
- D. $\frac{1}{4}$
- E. $\frac{1}{2}$

7. In set theory, an empty set is represented with

- A.
- B. $\{ \}$
- C. $\{.\}$
- D. A and B
- E. All of the above

8. If $A = \{x \in \mathbb{N} : x^2 - 3x + 2 = 0\}$, $B = \{x \in \mathbb{N} : x(x - 3) = 0\}$, $A \cap B$ is

- A. $\{3\}$
- B. $\{12\}$
- C. $\{ \}$
- D. $\{2\}$
- E. $\{1, 2, 3\}$

9. What value of x satisfy the inequality $2x + 2 < 5$.

- A. $x < 1$
- B. $x < 1$
- C. $x > 1$
- D. $x > 1$
- E. $0 < x < 1$

10. If the 39th term of an arithmetic progression is 141, what is the first term if the common difference is 2

- A. 65
- B. 141
- C. 39
- D. None of the above
- E. All of the above

11. The sum to infinity of the series $1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} \dots$ is

- A. 1
- B. 2
- C. 3
- D. $\frac{1}{2}$
- E. $\frac{1}{3}$

12. The value of $\sin 300^\circ$ is

- A. 3
- B. $\frac{1}{2}$
- C. $-\frac{1}{2}$
- D. 3
- E. 1

13. How many sides has a regular polygon whose interior angle is 144° ?

- A. 12
- B. 9
- C. 8
- D. 10
- E. 11

14. Simplify $27^{2/3} \times 32^{2/5}$

- A. 36
- B. 9
- C. 4
- D. $\frac{1}{36}$
- E. None of the above

15. The equation $n^2 - 16n + 64 = 0$ has

- A. equal roots
- B. imaginary roots
- C. Distinct roots
- D. No roots
- E. None of the above

16. The values of C for which $2c^2 - 11c + 12 = 0$ are

- A. 4, $\frac{1}{2}$
- B. -4, $\frac{3}{2}$
- C. 4, $-\frac{3}{2}$
- D. -4, $-\frac{3}{2}$
- E. 4, 3

17. Which of the following is not a surd
 A. 2 B. 3 C. 24 D. 9 E. None of the above
18. Simplify $4^5 - 1$
 A. $5 - 1$ B. $5 + 1$ C. $1 - 5$ D. 4 E. $4/5$
19. Solve $7/8$ of $2\frac{1}{4} + (25/16 -$
20. For what values of x and y are $x + y = 2$ and $2x - y = 7$?
 A. $x = 3, y = -1$
 B. $x = -3, y = 1$
 C. $x = -3, y = -1$
 D. $x = 1, y = 3$
 E. $x = -1, y = 3$

Use the following data to answer: questions 21 – 25

6,0,1,2,6,8,2,1,5,1,4,9,6,1,5,6,5,6,8,6,6

21. The range of the values is
 A. 8 B. 9 C. 6 D. 7 E. 5
22. The sum of the numbers and the number of items are
 A. 85, 20 B. 85, 19 C. 80, 20 D. 95, 21 E. 80,19
23. The mean value is
 A. 4.47 B. 4.0 C. 4.25 D. 4.05 E. 4.21
24. The medium mark is
 A. 6 B. 5 C. 7 D. 8 E. 4
25. The mode of the marks is
 A. 7 B. 6 C. 8 D. 5 E. 4

MATHEMATICS 2007/2008 SOLUTIONS/ANSWERS

11. $S = a = 1 = 2 = B$
 $1 - r = 1 - \frac{1}{2}$
12. $\sin 3000 = \sin (3600 - 600)$
 $= \sin 3600 \cdot \cos 600 - \sin 600 \cdot \cos 3600$
 $= (0 \times \frac{1}{2}) - (\frac{3}{2} \times 1) = -\frac{3}{2} = B$
13. For a regular polygon: internal angle + external angle = 1800

- $144 + x = 1800$
 27
 $X = 1800 - 1440 = 360$
 No of sides of regular polygon = $360/36 = 10$ sides = D
14. $272/3 \times 322/3 = (33)2/3 \times (25)2/5$
 $= 32 \times 22 = 36 = A$
15. $n^2 - 16n + 64 = n^2 - 8n - 8n + 64$
 $n = 8$ (twice) – Equal roots = A
16. $2c^2 - 11c + 12 = 0$ Using the quadratic equation on
 $-(-11) \pm \sqrt{(-11)^2 - 4(2)(12)} = \frac{11 \pm \sqrt{121 - 96}}{2(2)} = \frac{11 \pm 5}{4}$
 $X = 11/4$ or $25/4 = 11/4 + 5/4$ $x = 4$ or $x = 3/2 = A$
17. $9 = 3$ is no surd = D
18. $4 \times 5 + 1 = 4 \times 5 + 1 = 4 \times 5 + 1$
 $5 - 1 \times 5 + 1 \times 5 - 1 \times 4$
 $= 5 + 1 = B$
19. $7/8 \times 9/4 = (37/16 - 17/24)$
 $7/8 \times 9/4 = 77/48 = 7/8 \times 9/4 \times 48/77 = 15/22 = B$
20. $x + y = 2$ ZZZZZZ.. (1)
 $2x - y = 7$ ZZZZZZ.. (2) Subtracting (1) from (2)
 $3x = 9$ $x = 9/3 = 3$
 Substituting the value of x into (1) we have
 $3 + y = 2$ hence $y = -1 = A$
21. Range = highest value – lowest value
 $9 - 0 = 9 = B$
22. (94, 21) = D
23. $94/21 = 4.47 = A$
24. The Median is obtained by rearranging to order and then picking the middle term.
 0, 1, 1, 1, 1, 2, 2, 4, 5, 5, 5, 6, 6, 6, 6, 6, 8, 8, 9
 The median is 5 = B
25. The mode is the most occurring value is 6 = B

ENGLISH 2007/2008

In the flowing questions choose the expression or word which best completes each sentence.

1. The student who went home without an exit has apologized ____ his misconduct.
A. On
B. At
C. About
D. For
E. It
2. The man has atoned ____ his sins.
A. Upon
B. On
C. For
D. At
E. Against
3. I am ____ seeing your family.
A. Forward to
B. Ahead to
C. Forward on
D. Ahead to
E. At
4. These folk tales have been landed ____ from generation to generation
A. In
B. At
C. Over
D. Over
E. Eat
5. The young lovers first met _____ the University of Ibadan Night dancer
A. In
B. At
C. Inside
D. In course of
E. From
6. I have not seen my house master _____ the beginning of this session
A. From
B. In

- C. For
- D. Since
- E. Concerning

7. Get _____ the untimely death of his son

- A. Off
- B. By
- C. Through
- D. Over
- E. Eat

8. If you keep playing with this door handle, it will get ____

- A. lose
- B. Loose
- C. lost
- D. Loosing
- E. Renewed

In each of the following questions, choose the option nearest in meaning to the underlined word.

9. She was dressed in a gorgeous costume

- A. Richly coloured
- B. Loose
- C. Badly sewn
- D. Tight fitting
- E. Bad Colour

10. Obi's reaction is too subtle to be understood

- A. Violet
- B. Real
- C. Clever
- D. Wild
- E. Cold

11. Many people are often deceived by superficial knowledge

- A. Cheap
- B. Shallow
- C. Attractive
- D. Penetrating

E. All of the above

12. His subjects rejoiced in the end of his tyrannical rule

- A. Cruel
- B. Long
- C. Just
- D. Peaceful
- E. Ancient

13. Danquah is a very versatile scholar things.

- A. Dull
- B. Clever at his special field
- C. Interested in many different things
- D. Slow
- E. lazy

14. Nigerian sailors are very virile

- A. Strong and manly
- B. Vindictive
- C. Virtuous
- D. Vicious and cunning
- E. Friendly

15. I have to wade through that stream

- A. Walk hurriedly
- B. Swain
- C. Toddle
- D. Walk with difficulty
- E. Walk

16. His rise to fame was **meteoric**

- A. Well deserved
- B. Very gradual
- C. Very swift
- D. Too slow
- E. All of the above

17. Martha came late this morning but she gave **plausible** excuse

- A. Reasonable

- B. Very interesting
- C. Detailed
- D. Pathetic
- E. Stupid

In each of the following questions choose the option opposite in meaning to the underlined word

18. A tentative date was given

- A. definite
- B. A provisional
- C. An amicable
- D. Convincing
- E. None of the

19. Obi was the hero of the story

- A. Assassin
- B. Villain
- C. Devil
- D. Criminate
- E. Goliath

20. The man drew a sword as people congregated round him

- A. Fled from
- B. Praised
- C. Gathered round
- D. Mobbed
- E. Bent

21. The point you have made is quite apt

- A. Helpful
- B. Irrelevant
- C. Illogical
- D. Insensitive
- E. All of the above

22. That little boy has become quite chubby

- A. Intelligent.
- B. Tall
- C. Thin

- D. Huge
- E. Large

23. The action was premeditated

- A. Unplanned
- B. Unnecessary
- C. Catastrophic
- D. Uncoordinated
- E. Good

24. The boy made flippant remark

- A. serious
- B. well-meant
- C. A correct
- D. An expected
- E. Yes

25. The weather is getting warmer, so the ice should thaw soon

- A. Frost
- B. Freeze
- C. Melt
- D. Escape
- E. Run

ENGLISH 2007/2008 SOLUTIONS/ANSWERS

1. = D 2. = C 3. = A 4. = B
5. = B 6. = D 7. = D 8. = B

9. Gorgeous: very beautiful and attractive; giving pleasure and enjoyment (2) of colours clothes etc) with very deep colours: impressive ANS Richly coloured = A

10. Subtle: not very noticeable or obvious (2) (of a person or their behaving in a clever way and using indirect methods, in order to achieve something ANS Clever = C

11. Superficial: ANS shallow = B

12. Tyrannical: using power or authority over people in an unfair and cruel way ANS cruel = A

13. Versatile = C

14. Virile: strong and full of energy = A

15. Wade: to walk with an effort through something especially water or mud = D
16. Meteoric: achieving success very quickly = C
17. Plausible: reasonable and likely to be true = A
18. Tentative: not definite or certain because you may want to change it later
ANS Definite = A
19. Hero: ANS villain = B
20. Congregate: to come together in a group ANS gathered round = C
21. Apt: suitable or appropriate in the circumstances ANS irrelevant = B
22. Chubby: slightly fat in a way that people usually find attractive ANS thin = C
23. Premeditated: planned in advance ANS unplanned = A
24. Flippant: showing that you do not take something as seriously as other people think you should ANS serious = A
25. Thaw: to turn back into water after being frozen ANS freeze = B

CHEMISTRY 2007/2008

1. Rhombic sulphur, monoclinic sulphur and amorphous sulphur are:
- A. Isomers
 - B. Isotopes
 - C. Allotropes
 - D. Monomers
 - E. Polymorphs
2. Laughing gas is
- A. nitrogen (II) oxide
 - B. Nitrogen (I) oxide
 - C. Nitrogen (III) oxide
 - D. Nitrogen (V) oxide
 - E. Ammonia
3. The correct name of the compound $\text{CH}_3\text{CH}_2\text{CONH}_2$ is
- A. N-methyl-propionyl amide
 - B. Propionyl amide
 - C. Methyl propanamide
 - D. Methyl propionylamide
 - E. Propionylmethamide

4. The name pentanone is not specific and proper because it can refer to
A. 2-pentanone and 3-pentanone
B. 1-pentanone and 5-pentanone
C. Methyl propanone and propyl methanone
D. Methyl propanonne and methyl propyl butanone
5. The oxide that remains unchanged when heated with hydrogen is
A. CUO B. Fe₂O₃ C. PoO₂ D. ZnO E. Ag₂O
6. What is the most metallic element in the set
A. Na B. Ar C. P D. A E. Cl
7. What is the mole fraction of water in a solution prepared by mixing 12.5g of H₂O with 220g of acetone?
A. 0.817
B. 0.845
C. 0.183
D. 0.155
E. 0.205
8. A consequence of global warming is:
A. Air pollution
B. Wafer pollution
C. Increased humidity
D. Flooding
E. Little sunshine
9. Gunpowder is made from charcoal, Sulphur and Potassium trioxonitrate(V). The salt in the mixture perform[^] function of:
A. An oxidant
B. A reductant
C. A solvent,
D. A catalyst
E. Exploder
10. Which one of the following group II elements has the highest first ionization energy?
A. Be B. Mg C. Ca D. Sr E. B

11. In which one of the following compounds does sulphur have an oxidation number of +4?
- A. H_2SO_4
 - B. H_2S
 - C. SO_2
 - D. Na_2SO_3
 - E. $\text{H}_2\text{S}_2\text{O}_7$
12. Which one of the following does not show allotropy at room temperature and pressure?
- A. Nitrogen
 - B. Phosphorous
 - C. Oxygen
 - D. Carbon
 - E. Sulphur
13. Cellulose and starch can be classified as one of the following:
- A. Sugar
 - B. Sucrose
 - C. hydrocarbon
 - D. Carbohydrates
 - E. Fibres
14. What mass of anhydrous sodium trioxocarbonate (IV) is required to prepare 250cm^3 of 0.10M solution?
(Na = 23, C = 12, O = 16)
15. How many isomers can be formed from organic compounds with molecular formula $\text{C}_3\text{H}_8\text{O}$?
- A. 2 B. 3 C. 4 D. 5 E. 6
16. Which of the following is not a property of magnesium oxide?
- A. High melting point
 - B. Dissolution in polar solvent
 - C. Presence of ionic bonds
 - D. Possession of crystal lattice
 - E. low binding energy

17. Catalytic hydrogenation of alkenes produces compounds with general formula A. $C_nH_{2n+1}OH$.

- B. C_nH_{2n+1}
- C. C_nH_{2n+2}
- D. C_nH_{2n-1}
- E. $C_n(H_2O)_y$

18. Tetraoxosulphate (VI) acid is described as strong acid because it is highly

- A. Corrosive
- B. Concentrated
- C. Reactive
- D. Soluble in water
- E. Ionized in water

19. Which of these metals. Mg, Fe, Pb, and Cu will dissolve in dilute HCl if air is blown through the solution?

- A. Mg and Fe only
- B. All the metals
- C. Mg, Fe and Cu
- D. Mg, Fe and Pb
- E. Mg, Pb and Cu. —

20. A correct electrochemical series can be obtained from K, Na, Ca, Al, Mg, Zn, Fe, Pb, H. Ag, Au by interchanging

- A. Zn and Fe
- B. Zn and Pb
- C. Al and Mg
- D. Au and Ag
- E. None of the above. '

21. The oxidation of $CH_3CH_2(CH_3)OH$ gives:

- A. 2-butanone
- E. Methyl butane
- B. 2-butanal
- C. Butane
- D. Butanoic

22. The solubility of alkanols in water is due to:

- A. Their ionic character

- B. Their covalent nature
- C. Ability to form hydrogen bond
- D. Their low boiling points
- E. Their low dielectric com

23. The gas that is not associated with global warming is

- A. CO₂
- B. H₂
- C. CH₄
- D. SO
- E. SO₄

24. In which order are the following salts sensitive to light:

- A. AgBr>AgCl>AgI
- B. AgCl>AgI>AgBr
- C. AgI>AgCl>AgBr
- D. AgCl>AgBr>AgI
- E. AgBr>AgI>AgCl

25. A phenomenon where an element exists in different forms in the same physical state known as:

- A. Isomerism
- B. Amorphism
- C. Isotopy
- D. Allotropy
- E. Enantiomerism

CHEMISTRY 2007/2008 ANSWERS

- | | | | | |
|------|-------|-------|-------|-------|
| 1. C | 7. D | 13. D | 19. A | 25. D |
| 2. B | 8. D | 14. A | 20. C | |
| 3. A | 9. E | 15. A | 21. A | |
| 4. A | 10. A | 16. E | 22. C | |
| 5. D | 11. C | 17. C | 23. A | |
| 6. A | 12. A | 18. E | 24. D | |

PHYSICS 2007/2008

1. A piece of rubber 10cm long stretches 6mm when a load of 100N is hung from it. What is the strain?

- A. 6×10^2
- B. 60
- C. 6
- D. 0.6
- E. 2×10^{-2}

2. Which of the following does not cause a reduction of the surface tension of water?

- A. Soap solution
- B. Alcohol
- C. Camphor
- D. Grease
- E. Solvent

3. The amount of heat required to raise the temperature of a body is

- A. Thermal capacity
- B. Thermal energy
- C. Specific heat capacity
- D. Heat lost
- E. heat gain

4. Water shows anomalous behaviour

- A. below 100C
- B. At exactly 4°C
- C. between 4°C and 100°C
- D. Between 0°C and 4°C
- E. Above 1000C

5. Which of the following phenomena cannot be explained by the molecular theory of matter?

- A. Radiation
- B. Conduction
- C. Convection
- D. Evaporation
- E. Saturation

6. A gas occupies a volume of 300cm^3 at a temperature of 27°C . What is its volume at 540°C , when the pressure is constant?

- A. 150cm^3 .
- B. 273cm^3
- C. 327cm^3
- D. 600cm^3
- E. 125cm^3

7. A man clapping his hands at regular intervals observes that the echo of a clap coincides with the next clap. If the reflecting cliff is 160m away and the speed of the sound is 320m s^{-1} , what is the frequency of the clapping?

- A. 2Hz
- B. 4Hz
- C. 8Hz
- D. 1Hz
- E. 12Hz

8. Which of the following properties is/are common to all waves? I Diffraction
II Refraction III Frequency

- A. II only
- B. I and II only
- C. I, II and III only
- D. I and III only
- E. II only

9. Which of the following factors affects the speed of sound in air?

I Temperature II Pressure III Frequency

- A. II only
- B. I and II only
- C. I only
- D. II and III only
- E. III only :

10. When white light is dispersed by a spectrometer, the component having the shortest wavelength is

- A. Orange B. Green C. Red D. Violet E. Black

11. Shadows and eclipses result from the

- A. Refraction of light
- B. Diffraction of light

- C. Rectilinear propagation light
- D. Reflection of light
- E. Above tight

12. Which of the following media allow the transmission of sound waves through them: I Air II Liquid III Solid.

- A. I, II and III
- B. I and II only
- C. I and III only
- D. II and III only
- E. III only

13. The power dissipated in an a.c circuit with an r.m.s of 5A, r.m.s voltage of 10V and a phase angle of 60° is

- A. SOW
- B. 120W
- C. 125W
- D. 25W
- E. 12W

14. A light of energy 5eV falls on a metal and the electrons with a maximum kinetic energy of 2eV are ejected. The work fraction of the metal is

- A. 0.4eV
- B. 7.0Ev
- C. 2.5eV
- D. 3.0eV
- E. 1.0eV

15. In semiconductors, the carriers of current at temperature are

- A. Electrons only
- B. Electrons and holes
- C. Holes only
- D. Electrons and ions
- E. Ions only

16. The temperature at which water vapour present in the air saturates the air and begins to condense is known as

- A. Boiling point
- B. Melting point
- C. Dew point

- D. Triple point
- E. Violet light

17. Which of the following pairs is not part of the electromagnetic spectrum?

I Radio wave II Beta waves III Gamma rays IV Alpha rays

- A. I and II
- B. II and IV
- C. III and IV
- D. I and III
- E. all of the above

18. A wave of frequency 10Hz forms a stationary wave pattern in a medium where the velocity is 20cmS⁻¹. The distance between the adjacent modes is

- A. 15cm
- B. 1.0cm
- C. 2.0cm
- D. 5.0cm
- E. 6.0cm

19. The length of a displaced pendulum bob which passes its lowest point twice every second, assuming $g = 10\text{ms}^{-2}$ is

- A. 0.25m
- B. 0.45m
- C. 0.58m
- D. 1.00m
- E. 1.2m

20. The inner diameter of a small test tube can be measured accurately using a

- A. micrometer screw gauge
- B. Pair of dividers
- C. Meter rule
- D. Pair of vernier calipers
- E. meter screw

21. An object is projected with a velocity of 80ms⁻¹ at an angle of 30° to the horizontal. The maximum height reached, assuming $g = 10\text{m/s}^2$, is

- A. 20m
- B. 80m
- C. 160m

- D. 320m
- E. 40m

22. A cone in an unstable equilibrium has its potential energy,

- A. Decreased
- B. Increased
- C. Oscillating
- D. Unchanged
- E. Undulating

23. A car of mass 800kg attains a speed of 25ms⁻¹ in 20 seconds. The power developed in the engine is

- A. 12.5KW
- B. 25.0KW
- C. 1.25MW
- D. 2.50MV
- E. 1MV

24. When a ship sails from salt water into fresh water, fraction of its volume above the water surface will

- A. Decrease
- B. Remain the same
- C. Increase
- D. Increase then decrease
- E. All of the above

25. A machine gun with a mass of 5kg fires a 50g bullet at a speed of 100ms⁻¹. The recoil speed of the machine gun is

- A. 0.5m/s
- B. 3.5m/s
- C. 1m/s
- D. 2m/s
- E. 4m/s

PHYSICS 2007/2008 SOLUTIONS/ANSWERS

- 1. Strain = elongation = $L = 6\text{mm}$
Original –length $L_0 (10 \times 10)$
 $28 = 0.06 = 6 \times 10^{-2} = A$
- 2. D

3. C
4. D
5. A
6. Since pressure is constant: $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
 $300 = \frac{V_2 \times 327}{300}$ thus $V_2 = \frac{300 \times 300}{327} = 277.37 = C$
7. $v = 320 \text{ m/s}$
 The echo travels: distance $\times 2 = 2 \times 160$ and $v = 2f$
 $320 = 2 \times f \times 160$ thus $f = \frac{320}{2 \times 160} = 1 \text{ Hz} = D$
8. = C
9. C
10. D
11. C
12. A
13. $P = VI \cos \theta$
 $10 \times 5 \times \cos 60 = 25 \text{ W} = D$
14. $E_k = h\nu - W$; $W = E_k - h\nu = 5 - 2 = 3 \text{ eV} = D$
15. B
16. C
17. B
18. $v = f \lambda$ and $\lambda = \frac{v}{f} = \frac{20}{10} = 2.0 \text{ cm}$
 But distance between adjacent nodes is $\frac{2.0}{2} = 1.0 \text{ cm} = B$
19. $T = 2\pi \sqrt{\frac{l}{g}}$
 $(\frac{T}{2})^2 = \frac{l}{g}$ $(\frac{T}{2})^2 = \frac{l}{10}$ and $l = 0.25 \text{ m} = A$
20. = D
21. $H = \frac{1}{2} U^2 \sin^2 \theta = 802 (\sin 300)^2$
 $2g \times 10 = 80 \text{ m} = B$
22. = A decreased
23. B
24. A
25. STV

COMMERCE AND ACCOUNT 2007/2008

1. The bad debt account is closed by transferring the balance to
 - (a) credit of capital account
 - (b) debt of profit and loss
 - (c) debit of the debtor' personal account
 - (d) credit of profit and loss account

2. A credit note from a supplier would first e entered in the
 - (a) purchase book
 - (b) sales book
 - (c) returns inward book
 - (d) return outward

3. Where a proprietor withdraws cash from the business for his private use, the entries would be
 - (a) credit cash account, debit proprietor's account
 - (b) credit proprietor' account, debit cash account
 - (c) credit proprietor' account, debit drawing account
 - (d) credit cash account, debit drawing account

4. Which of the following is a long term liability?
 - (a) bank overdraft
 - (b) bank deposit
 - (c) goodwill
 - (d) debenture

5. Which of the following is a function of the sales journal?
 - (a) it records both cash and credit sales
 - (b) it shows the balance in the sales account
 - (c) it records credit sales only
 - (d) it shows the aggregate balances in the debtor's lodge

6. Which of the following is the most liquid?
 - (a) stock
 - (b) debtor
 - (c) machinery
 - (d) loose tools

7. A debt balance on a bank statement indicate, that the

- (a) bank is a debtor to the customer
 - (b) customer is a debtor to the bank
 - (c) bank is not a debtor to the customer
 - (d) customer is not a debtor to the bank
8. At the end of a trading period, wages due will be
- (a) debited to trading account and shown as a current liability
 - (b) credited to profit and loss account and shown as a current liability
 - (c) credit to trading account and shown as a current asset
 - (d) debited to profit and loss account and shown as a current liability
9. Which of the following is not an asset?
- (a) goodwill
 - (b) general reserves
 - (c) debtors
 - (d) prepaying
10. A company's profit and loss account for a period is prepared by
- (a) deducting total expenses from total revenues
 - (b) deducting total cash payments from total revenue
 - (c) deducting all outflows of funds from all inflow of funds
 - (d) balancing its receipts and payment
11. The journal is a book
- (a) of prime entry into which all transactions are entered before posting
 - (b) kept by journalists for reporting events
 - (c) of prime entry into which posting are made from the ledger
 - (d) of analyzing major events
12. The receipts and payments of a club is the same as
- (a) profit and loss account
 - (b) trading account
 - (c) cash book
 - (d) control account
13. In a balance sheet, bank overdraft is
- (a) current asset
 - (b) fixed asset
 - (c) long term liability

(d) current liability

14. Money set aside for the servicing of businesses overheads is referred to

- (a) capital expenditure
- (b) capital receipt
- (c) revenue expenditure
- (d) revenue receipts

15. Money spent on buying plots of land by a firm is

- (a) capital expenditure
- (b) capital receipt
- (c) revenue expenditure
- (d) revenue receipts

16. Prepaid expenses is an item in

- (a) fixed assets
- (b) current assets
- (c) current liability
- (d) long-term loan

17. Final accounts consists of (i) Cash book (ii) Trading account (iii) Profit and loss account (iv) balance sheet

- (a) i only
- (b) i and ii only
- (c) iii and iv only
- (d) ii and iii only

18. Trading account is prepaid to ascertain

- (a) gross profit
- (b) net profit
- (c) gross profit or gross loss
- (d) net profit or not loss

19. A statement that shows the financial position of a business is

- (a) profit and loss account
- (b) trading account
- (c) statement of income
- (d) balance sheet

20. Prepaid wages is an item in the (i) balance sheet (ii) profit and loss account (iii) Trading account (iv) cash account
(a) I, ii and iii only
(b) I and iii only
(c) I and ii only
(d) iii and iv only
21. Partnership is a business owned and controlled by
(a) two member
(b) two to ten persons
(c) two to twenty persons
(d) two to five persons
22. If a machine cost N2000.00 and its estimated life span is 10 years and its residuals is N500. The yearly value of depreciation is
(a) N250.00
(b) N180.00
(c) N150.00
(d) N130.00
23. Bad debt is also known as
(a) unpaid debt
(b) irrecoverable debt
(c) payable debt
(d) current debt
24. Bad debt is an item in
(a) Trading, profit and loss account
(b) consignment account
(c) balance sheet
(d) Receipt and expenditure account
25. An association of the principal business concerns in an area is referred to as
(a) Article of association
(b) Memorandum of association
(c) Chamber of commerce
(d) Chamber of companies

COMMERCE AND ACCOUNT 2007/2008 ANSWERS

1. C	2D	3A	4D	5C	6A
7A	8D	9B	10 -	11A	12C
13D	14B	15A	16C	17D	18C
19B	20B	21C	22C	23B	24B
25C					

**DELTA STATE UNIVERSITY, POST-UTME SCREENING PAST
QUESTIONS AND SOLUTIONS**

**ANSWER ALL QUESTION RELEVANT TO YOU:
SHADE THE ANSWER SHEET AS APPROPRIATE WITH HB PENCIL
ONLY**

ECONOMICS 2008/2009

1. Which of the following best describes a perfectly inelastic demand function?
 - A. the quantity demanded is insensitive to changes in price
 - B. price is insensitive to changes in quantity demanded
 - C. price and quantity demanded changes by some percentage
 - D. the demand function is horizontal
2. Which of the following statement is true?
 - A. a variable defined over a period of time is called a stock variable
 - B. microeconomics analyses interaction in the economy as a whole
 - C. a fall in the price of garri will shift its demand right ward
 - D. the market demand curve for commodity will shift to the right if the price of its substitute falls.
3. Which of the following statement is false?
 - A. Ends in Economics are the resources with which it can attain our goals
 - B. Tables and graphs are focus of arithmetic technique used in economics to present ideas pictorially
 - C. Three basic questions every economy must answer are what to produce, how to produce and for whom to produce.
 - D. A and B. are substitutes if the supply of A increases, the price of B will tend to increase
4. Which of the following statement is false?
 - A. An economy where economic decision making is shared by individuals and government is known as mixed economy
 - B. Given that apples and pears are substitutes, if the price of apples falls, the price of pears falls
 - C. an elasticity of demand measured between two points on a demand curve is called a point elasticity of demand
 - D. A and B are complements, if the supply of A decrease, the price of B will tend to fall

5. Which of the following statement is true?
- A. if cross price elasticity of demand between two goods, A and B is negative, we can conclude that A and B are substitutes.
 - B. the co-efficient of elasticity of demand is measured as a ratio of change in quantity demanded to change in price
 - C. a market will be in disequilibrium when demand is not equal to supply
 - D. all of the above
6. If the equation relating consumption (C) to income (Y) is $C = N7,500 + 0.2Y$, then
- A. consumption is inversely related to income
 - B. consumption is the independent variable and income is the dependent variable
 - C. if income is N15,000, the consumption is N10,500
 - D. if consumption is N30,000. then income is N10,000
7. The reason for the law of demand can best be explained in terms of:
- A. supply
 - B. complementary goods
 - C. the rationing function of prices
 - D. diminishing marginal utility
8. The demand curve for the pure monopolists is
- A. perfectly price elastic
 - B. perfectly price inelastic
 - C. downsloping
 - D. upsloping
9. Which of the following statement is correct? The Central Bank
- A. prints the country's currency
 - B. handles the collection of debts owed to the government
 - C. lends money to businessmen
 - D. issues the country's currency
10. Which of the following is not a basic assumption underlying the theory of consumer behaviour
- A. consumers prefer more to less
 - B. consumers are rational
 - C. consumers are willing to make choices

- D. consumers make consistent choices
11. Which of the following equations is the correct one about Net National Product (NNP?)
- A. $NNP = \text{Gross National Product (GNP)} - \text{Depreciation}$
 - B. $NNP = \text{Gross National Product (GNP)} - \text{Net income from abroad}$
 - C. $NNP = \text{Gross Domestic Product (GDP)} - \text{Depreciation}$
 - D. $NNP = \text{Gross Domestic Product (GDP)} - \text{Net income}$
12. Money market differs from capital market in that it
- A. deals with short-term loans while capital market deals with long term funds
 - B. deals with money only while capital market deals with capital as well
 - C. is limited in scope while capital market does not
 - D. uses interest rates while capital market does not
13. A production possibility curve shows
- A. how much of the resources of society are used to produce a particular commodity
 - B. the rate of inflation
 - C. the rate of unemployment in the economy
 - D. the various combinations of two commodities that can be produced
14. The situation in which the total amount of government spending exceeds total revenues is called a
- A. balanced budget
 - B. budget surplus
 - C. budget deficit
 - D. fiscal surplus
15. The maximization of profit tends to be the driving force in the economic decision making of:
- A. entrepreneurs
 - B. workers
 - C. consumers
 - D. legislators
16. Age distribution table

The table above shows the age distribution of a town in Nigeria. What is the dependency ratio of the town?

- A. 3.2
- B. 2:1
- C. 5.2
- D. 4:3

17. Which of these factors does not cause a change in demand?

- A. income
- B. population
- C. price of other commodities
- D. price of the commodity itself

18. When the demand for a commodity is inelastic, total revenue will fall if

- A. price is increased
- B. price remain constant
- C. price is reduced
- D. the commodity is a luxury

19. An economic system in which most capital goods are owned by individuals and private firms is known as

- A. planned economy
- B. capitalist economy
- C. mixed economy
- D. socialist economies from abroad

20. Which of the following is the most liquid asset to commercial bank?

- A. cash
- B. money at call
- C. stocks and shares
- D. Treasury Bills

ECONOMICS 2008/2009 ANSWERS

1.A	6.C	11.A	16.A	21.D	26.A
2.A	7.D	12.A	17.D	22.B	27.A
3.A	8.C	13.D	18.C	23.C	
4.B	9.D	14.C	19.B	24.C	
5.A	10.A	15.A	20.A	25.C	

CHEMISTRY 2008/2009

1. Homologous pairs of chromosomes separate during
 - (A) cytolysis
 - (B) cleavage
 - (C) mitosis
 - (D) meiosis

2. An example of caryopsis is
 - (A) guava
 - (B) maize grain
 - (C) coconut
 - (D) tomato

3. The response of plants to external stimuli in a non-directional manner is known as
 - (A) tactic movement
 - (B) phototropism
 - (C) geotropism
 - (D) nastic movement

4. The most important hormone that includes the ripening of fruit is
 - (A) ethylene
 - (B) indole acetic acid
 - (C) gibberallin
 - (D) cytokinin

5. One example of fossil fuels is
 - (A) limestone
 - (B) coral
 - (C) coal
 - (D) firewood

6. The most effective method of dealing with on-degradable pollutants is by
 - (A) dumping
 - (B) recycling
 - (C) incineration
 - (D) burying

7. Mycorrhiza is an association between fungi and

- (A) protozoan
- (B) roots of highest plants
- (C) bacteria
- (D) filamentous algae

8. A limiting factor in a plant population near a chemical factory is likely to be

- (A) light
- (B) humidity
- (C) wind
- (D) pH

9. Soil fertility can best be conserved and renewed by the activities of

- (A) earthworms
- (B) man
- (C) rodents
- (D) microbes

10. The pioneer organisms in ecological succession are usually the

- (A) mosses
- (B) ilohens
- (C) ferns
- (D) algae

11. The type of reproduction that leads to variant in animal and plant populations is

- (A) budding
- (B) sexual
- (C) vegetative
- (D) asexual

12. An insect with a mandibulate mouth part will obtain its food by

- (A) biting and chewing
- (B) chewing and sucking
- (C) chewing
- (D) sucking

13. Spines and shells on animals are adaptations of

- (A) camouflage
- (B) chemical defence

- (C) physical defence
- (D) mimicry

14. The rods in the retina of the eye are examples of

- (A) cells
- (B) tissues
- (C) organs
- (D) systems

15. A plant-like feature in *Euglena* is the

- (A) pellicle
- (B) pigment spot
- (C) large spot
- (D) gullet

16. The larval stage of mosquito is called

- (A) wriggler
- (B) grub
- (C) maggot
- (D) caterpillar

17. A peculiar characteristic of mammals is that they

- (A) have teeth
- (B) are warm-blooded
- (C) have lungs
- (D) have sebaceous glands

18. The gall bladder of a mammal has a duct connected to the (A) liver

- (B) duodenum
- (C) small intestine
- (D) pancreas

19. Rodent gnaw in food with their

- (A) molar teeth
- (B) strong jaws
- (C) flat-ridged teeth
- (D) chisel-like front teeth

20. An example of a radically symmetrical organism

- (A) planarian
- (B) hydra
- (C) tapeworm
- (D) roundworm

CHEMISTRY 2008/2009 ANSWERS

- | | | | |
|------|-------|-------|-------|
| 1. C | 7. B | 13. C | 19. D |
| 2. B | 8. D | 14. A | 20. B |
| 3. D | 9. A | 15. B | |
| 4. A | 10. B | 16. B | |
| 5. C | 11. B | 17. B | |
| 6. B | 12. A | 18. D | |

MATHEMATICS 2008/2009

1. Find the slope of the curve $y = 2x^2 + 5x - 3$ at (1,4)
(A) 4 (B) 3 (C) 7 (D) 9
2. Determine the maximum value of $y = 3x - x^2$
(A) 0 (B) 2 (C) 4 (D) 6
3. By how much will the mean of 30, 56, 31, 55, 43 and 44 less than the median
(A) 0.75 (B) 0.50 (C) 0.33 (D) 0.17
4. The range of 4, 3, 11, 9, 6, 15, 19, 23, 27, 24, 21 and 61 is
(A) 16 (B) 21 (C) 23 (D) 24
5. The mean of the numbers 3, 6, 4, x and 7 is 5, Find the standard deviation
(A) 2 (B) 5 (C) 12 (D) 10
6. Find the remainder when $3x^3 + 5x^2 - 11x + 4$ is divided by $x + 3$.
(A) 4 (B) 1 (C) -1 (D) -4
7. What are the integral values of x which satisfy the inequality $-1 < 3 - 2x < 5$?
(A) 2, 1, 0, -1 (B) -1, 0, 1, 2 (C) -1, 0, 1 (D) 0, 1, 2.
8. Find the derivative of $(2 + 3x)(1 - x)$ with respect to x.
(A) $6x - 1$ (B) $1 - 6x$ (C) 6 (D) -3

9. Find the derivative of the function $y = 2x^2(2x - 1)$ at the point $x = -1$
(A) -6 (B) -4 (C) 16 (D) 18
10. Find the mean deviation of 1, 2, 3 and 4
(A) 1.0 (B) 1.5 (C) 2.0 (D) 2.5
11. Find the value of t if the standard deviation of $2t, 3t, 4t, 5t$ and $6t$ is 2.
(A) 1 (B) 2 (C) 3 (D) 4
12. In how many ways can 6 coloured chalks be arranged if are of the same colour?
(A) 60 (B) 120 (C) 240 (D) 360
13. A final examination requires that a student answers any 4 out of 6 questions. In how many ways can this be done?
(A) 15 (B) 20 (C) 30 (D) 45
14. If the mean of five consecutive integers is 30, find the largest of the numbers
(A) 28 (B) 30 (C) 32 (D) 34
15. A bag contains 5 black, 4 white and x red marbles. If the probability of picking a red marble is $\frac{2}{5}$ find the value of x
(A) 8 (B) 10 (C) 4 (D) 6.
16. For what values of n is $n+1 C_3 = 4(n C_3)$?
(A) 6 (B) 5 (C) 4 (D) 3
17. Find the roots of $x^3 - 2x^2 - 5x + 6 = 0$
(A) 1,2, -3 (B) -1,2, 3 (C) -1,2,-3 (D) 1,-2, 3
18. Find the value of k if the expression $kx^3 + x^2 - 5x - 2$ leaves a remainder 2 when divided by $2x + 1$
(A) 10 (B) 8 (C) -10 (D) -8
19. If $y = x^2 - x - 12$ find the range of values of x for which $y > 0$.
(A) $x < -3$ or $x > 4$
(B) $x < -3$ or $x > 4$

- (C) $-3 < x < 4$
 (D) $-3 < x < 4$

20. How many terms of the series 3, -6, +12, -24 + ... are needed to make a total of 1 -28?

- (A) 12 (B) 10 (C) 9 (D) 8

MATHEMATICS 2008/2009 SOLUTIONS/ANSWERS

- From the equation of a straight line: $y = mx + c$ where m is the slope
 $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 0}{1 - 0} = 4$ and $dy = 4x + 5$
 $\frac{dy}{dx} = 4x + 5$
 when $x = 1$: $dy = 4(1) + 5 = 9 = D$
- $y = 3x^2 - x^3$ and $\frac{dy}{dx} = 6x - 3x^2$ we that at Max. value $\frac{dy}{dx} = 0$
 $6x - 3x^2 = 0$; $x(6 - 3x) = 0$ Hence either:
 $x = 0$ or $6 - 3x = 0$; $x = 2$ or $0 = B$
- The Mean $\bar{x}$ of the distribution = $\frac{\sum x}{N} = \frac{30 + 56 + 31 + 55 + 43 + 44}{6} = 43.17$
 The median, after rearranging the data; 30, 31, 43, 44, 55, 56
 Median $(\frac{n+1}{2}) = \frac{6+1}{2} = 3 \frac{1}{2}$ th position = $\frac{43 + 44}{2} = 43.5$
 Median - Mean = $43.5 - 43.17 = 0.33 = C$
- Range = $61 - 3 = 58$
- $\sum x_i = 3 + 6 + 4 + x + 7 = 20 + x$
 $N = 5$
 $\sum x_i^2 = 3^2 + 6^2 + 4^2 + x^2 + 7^2 = 25 + 20x + x^2$
 $(S.D)^2 = \frac{\sum x_i^2}{N} - (\bar{x})^2 = \frac{25 + 20x + x^2}{5} - (\frac{20+x}{5})^2 = 10$
 $(S.D)^2 = 2$ and $S.D = \sqrt{2} = A$
- From the remainder theorem; we take from $x + 3$ and factor out for when
 $x = -3$. $f(-3) = 3(-3)^3 + 5(-3)^2 - 11(-3) + 4 = 1$. You can solve this by LONG division but it is LONG OK? Remainder = 1 = B
- $-1 < 3 - 2x < 5$
 $-1 - 3 < -2x < 5 - 3$
 $-4 < -2x < 2$

$$X < 2 \quad x > -1$$

The integral values of x which satisfy the inequality can be obtained from the range above. They are $-1, 0, 1 = C$

$$8. y = (2 + 3x)(1 - x) = 2 - 2x + 3x - 3x^2$$

$$y = 2 + x - 3x^2 \quad dy = 1 - 6x = B$$

$$9. y = 2x^2 (2x - 1) = 4x^3 - 2x^2 \text{ and } dy = 12x^2 - 4x \text{ and when } x = -1$$

$$Dx$$

$$dy = 12(-1)^2 - 4(-1) = 12 + 4 = 16 = C$$

$$dx$$

$$10. \text{Mean} = x = x = 1 + 2 + 3 + 4 = 10 = 2.5$$

$$N = 4$$

$$\text{Mean deviation} = (x - \bar{x}) = (1 - 2.5) + (2 - 2.5) + (3 - 2.5) + (4 - 2.5) = 10$$

$$N = 4 = A$$

$$11. x = 2t + 3t + 4t + 5t = 20t$$

$$5 \times 5 = 4t$$

$$20$$

$$(S.D)^2 = (x - \bar{x})^2 = (2t - 4t)^2 + (3t - 4t)^2 + (4t - 4t)^2 + (5t - 4t)^2 = 10t^2$$

$$N = 5$$

$$(2) = 10t^2 = 2; \text{ cross multiplying we get that}$$

$$t^2 = 1 \text{ and } t = 1 = A$$

$$12. \text{Since 2 are of the same colour, we have:}$$

$$2 \times 6P_2 = 2 \times 6! = 2 \times 6! = 60 = A$$

$$(6-1)! = 4!$$

$$13. \text{Selection entails combination thus; } {}^6C_4 = 15 \text{ ways} = A$$

$$14. \text{Let the first integer of the numbers be } x:$$

$$X = x + (x + 1) + (x - 2) + (x - 3) + (x - 4) = 5x + 10$$

$$5 \times 5 = 30$$

$$5x + 10 = 150; 5x = 140 \text{ and } x = 28$$

$$\text{Hence largest of the numbers is } = x + 4 = 28 + 4 = 32 = C$$

$$15. n(\text{Black}) = 5, n(\text{White}) = 4, n(\text{Red}) = x; \text{ hence total} = x + 9 \text{ and } P(\text{red}) = x$$

$$x+9. \text{ From the question we have that; } P(\text{red}) = \frac{2}{5} \text{ thus from above}$$

$$x = \frac{2}{5}. 5x = 2(x + 9) = 2x + 18$$

$$x+9. 5x - 2x = 18 \text{ and } x = 6 = D$$

$$16. n+1C_3 = 4(nC_3) = (n+1)! = 4n!$$

$$3!(n+1-3)!3!(n-3)!$$

$$(n+1)(n)(n-1)(n-2)! = 4n(n-1)(n-2)(n-3)(n-4)!$$

$$3!(n-2)! \cdot 3!(n-3)(n-4)!$$

$$n(n+1)(n-1) = 4[n(n-1)(n-2)]$$

$$n-4n = -8-1 = -9. \text{ Hence } n = 3 = D$$

17. $f(1) = 13 - 2(1)^2 - 5(1) + 6 = 0$, thus $(x-1)$ is a factor and by long division we get

$$x^2 - x - 6.$$

$(x-1)(x^2 - x - 6)$ are factors thus we have

$$(x-1)(x-3)(x+2) = 0$$

$$X = 1, -2, 3 = D$$

18. $kx^3 + x^2 - 5x - 2$

$$2x + 1 = 0 \text{ thus } x = -\frac{1}{2}$$

$$F(-\frac{1}{2}) = k(-\frac{1}{2})^3 + (-\frac{1}{2})^2 - 5(-\frac{1}{2}) - 2 = 2 \Rightarrow -k + 1 + 5 - 2 = 2$$

$$8 - 4 = 2$$

simplifying further we have $-k = 16 - 2 - 20 + 16$

$$-k = 32 - 22 = 10$$

$$K = -10 = C$$

19. To find the range of x for which $y > 0$: i.e. $x^2 - x - 12 = 0$

$(x+3)(x-4) = 0$ thus $x = 3$ or -4 we can plot the inequality graph for this to obtain $x < -3$ and $x > 4 = B$

20. Sum of the series $= 1 - 28 = -255$

$$r = -6/3 = -2, a = 3$$

$$\text{sum} = a(1 - r^n) / (1 - r) = 3(1 - (-2)^n) / (1 - (-2)) = -255$$

$$1 = r^{n+1} + 2$$

$$-256 = 1 - (-2)^n : \text{ and } 256 = (-2)^n$$

$$(-28) = (-2)^n \text{ hence } n = 8 = D$$

ENGLISH 2008/2009

From the list of words choose the one that best completes each sentence from 1 to 15

1. You are driving - fast for my liking

(A) too (B) very (C) pretty (D) fairly

2. You have given me one orange - many

(A) very (B) so (C) too (D) more

3. The upholstery work doesn't go - the colour of the car

(A) after (B) by (C) with (D) for

4. I became depressed -hearing the news

(A) at (B) with (C) as (D) on

5. He was punished for failing -his duty as a prefect of the school

(A) on (B) about (C) with (D) in

6. Good discipline was instructed -the success achieved by the college

(A) for (B) to (C) in (D) with

7. It was quite dark in the room -we couldn't see

(A) so
(B) because
(C) through
(D) yet

8. He needed work so late -he

(A) does (B) needs (C) did (D) need

9. If I had known he would come, I -have gone to meet him

(A) may (B) will (C) should (D) must

10. He -thirty when I first met him

(A) must have been
(B) will have
(C) ought to be
(D) must have to be

11. Where is that brother of--now?

(A) you (B) yours (C) our (D) your

12. Someone told me where it was, but I can't remember –

(A) whom (B) it (C) who (D) didn't

13. I guessed it was going to rain— ?

(A) did I (B) was it (C) wasn't it (D) didn't I

14. Would you like the door — ?

- (A) close (B) closes (C) closed (D) closing

15. You shouldn't put off---the assignment

- (A) to do
(B) to have done
(C) having done
(D) to have been

Choose the word(s) that is/are nearly composite in meaning to the underlined word and which correctly fill the blank in the sentence. From 16 to 20

16. The **able-bodied** should take care of the ---

- (A) feeble
(B) weak-minded
(C) suffering
(D) soft-hearted

17. The chairman ordered him either to **withdraw** or to ---his allegations

- (A) affirm
(B) draw
(C) express
(D) complete

18. He shows plenty of **good-will** to his neighbours, but they bear nothing except -towards him

- (A) bad luck
(B) malice
(C) anger
(D) unhappiness

19. I **supported** you said but I -the way you said it

- (A) argued about
(B) objected to
(C) interfered with
(D) investigated

20. Though many of us were **poor** quite a few were

- (A) arrogant
(B) prodigal

- (C) affluent
- (D) luxurious

ENGLISH 2008/2009 ANSWERS

- 1. C 8. C 15. A
- 2. B 9. A
- 3. C 10. A
- 4. D 11. B
- 5. D 12. C
- 6. A 13. D
- 7. B 14. C

16. able-bodied: physical healthy, fit and strong in contract to somebody who is weak or disable ANS feeble = A

17. withdraw: to move back or away from a place or situation; to make somebody or something do this ANS affirm = A

18. good-will: friendly or helpful feelings towards other people or countries ANS malice = B

19. Supported: to help or encourage somebody or something by saying/showing that you agree with them/it ANS objected to = B

20. Poor: you should know the proper definition of poverty if you are a Nigerian.

Affluent: having a lot of money and a good standard of living = C

PHYSICS 2008/2009

1. The wavelength of the first overtone of a note in a closed pipe of length 33cm is

- (A) 44cm (B) 33cm (C) 22cm (D) 17cm

2. None-luminous objects can be seen because they

- (A) are polished
- (B) are near
- (C) reflect light
- (D) emit light

3. The correct unit of energy density is

- (A) $\text{kg m}^{-3}\text{s}^{-2}$
- (B) $\text{kg m}^{-1}\text{s}^{-2}$
- (C) $\text{kg m}^3\text{s}^{-2}$

(D) $\text{kg m}^2\text{s}^{-2}$

4. The motion of smoke particles from a chimney is typical of

- (A) oscillatory motion
- (B) rotational motion
- (C) circular motion
- (D) random motion

5. One of the properties of gamma rays is that they are (A) negatively charged

- (B) massive
- (C) neutral
- (D) positively charged

6. The process whereby the molecules of different substances moves randomly is called

- (A) surface tension
- (B) diffusion
- (C) capillarity
- (D) osmosis

7. The process whereby a liquid turns spontaneously into vapour is called

- (A) evaporation
- (B) regelation
- (C) boiling
- (D) sublimation

8. The velocity of sound in air will be doubled if its absolute temperature is

- (A) doubled
- (B) halved
- (C) constant
- (D) quadrupled

9. A thin converging lens has a power of 4.0 diopters, determine its focal length (A) 0.25m

- (B) 0.03m
- (C) 5.00m
- (D) 2.50m

10. An electric device is rated 2000W, 250V. The correct fuse rating of the device is
(A) 8A
(B) 9A
(C) 7A
(D) 6A
11. Satellite communication network makes use of
(A) infra-reds
(B) sound wave
(C) visible light
(D) radio waver
12. If two inductors of inductances 3H and 6H are arranged in series, the total inductance is
(A) 18.0H
(B) 9.0H
(C) 2.0H
(D) 0.5H
13. The current in a reverse-biased junction is due to
(A) electrons
(B) majority carriers
(C) holes
(D) minority carriers
14. In an a.c circuit that contains only a capacitor, the voltage lags behind the current by
(A) 90°
(B) 180°
(C) 600
(D) 300
15. The ray which causes gas molecules to glow is known as
(A) molecular ray
(B) gamma ray
(C) anode ray
(D) cathode ray

16. The charge carriers in gases are

- (A) ions only
- (B) electrons and holes
- (C) electrons only
- (D) electrons and ions

17. Which of the following materials is a conductor?

- (A) plastic
- (A) plastic
- (B) sodium
- (C) wax
- (D) glass

18. The instrument used for securing a large number of similar charges by induction is called

- (A) capacitor
- (B) electrophorous
- (C) electroscope
- (D) proof-plane.

19. The pitch of a sound note depends on

- (A) timbre
- (B) harmonics
- (C) quality
- (D) frequency

20. In which of the following material media would sound travel fastest?

- (A) water
- (B) oil
- (C) metal
- (D) gas

PHYSICS 2008/2009 SOLUTIONS/ANSWERS

1. $l = 3/4 \Rightarrow 3 = 4l = 4 \times 33 = 132$

$3l = 132$ hence $l = 44\text{cm} = A$

21

- | | | |
|------|------|------|
| 2. C | 5. C | 8. A |
| 3. D | 6. B | |
| 4. D | 7. A | |

9. The power of the lens is equal to the reciprocal of the focal length and is measured in dioptries when f is in meters.

$$P = 1/f \Rightarrow f = 1/P = 0.25\text{m} = A$$

10. $P = IV$ and $I = P/V$ $2000/250 = 8A = A$

11. D

12. Since it all acts as resistance; resistance in series $R = R_1 + R_2$

$$\text{Hence inductance} = 3H + 6H = 9H = B$$

13. Reverse-biased junction (.i.e P-semiconductors) have more holes than electrons but the holes are the majority carriers of current= C

14. V_c lags I_c by $90^\circ = A$

15. D 18. B

16. D 19. D

17. B 20. D

BIOLOGY 2008/2009

1. The shape of the s-orbital is

- (A) spherical
- (B) elliptical
- (C) spiral
- (D) circular

2. A carcinogenic substance is

- (A) asbestos dust
- (B) saw dust
- (C) nitrogen (II) oxide
- (D) carbon (II) oxide

3. In the electrolysis of brine the anode is

- (A) platinum
- (B) copper
- (C) zinc
- (D) carbon

4. Which of the following halides has the highest entropy value

- (A) HF
- (B) HCl
- (C) HBr
- (D) HI

5. The allotrope of carbon used in the decolourization of sugar is
(A) graphite
(B) soot
(C) charcoal
(D) lampblack
6. Sulphur (IV) oxide bleaches by
(A) reduction
(B) oxidation
(C) hydration
(D) absorption
7. Aluminum hydroxide is used in the dyeing industry as a
(A) salt
(B) dye
(C) mordant
(D) dispersant
8. An isomer of C_5H_{12} is
(A) butane
(B) 2-methylbutane
(C) 2-methylpropane
(D) 2-ethylbutane
9. Vulcanization involves the removal of
(A) a monomer
(B) the single bond
(C) the double bond
(D) a polymer
10. Phenolphthalein in acid solution is
(A) red
(B) orange
(C) colourless
(D) yellow
11. When iron is exposed to moisture and it rusts, the value of G for the reaction is

- (A) neutral
- (B) zero
- (C) positive
- (D) negative

12. A substance that is used as a ripening agent for fruits is

- (A) ethene
- (B) propene
- (C) methane
- (D) butane

13. The shape of the hydrocarbon compound CH_4 is

- (A) square plannar
- (B) plannar
- (C) linear
- (D) tetrahedral

14. Sugar is separated from its impurities by

- (A) precipitation
- (B) crystallization
- (C) distillation
- (D) evaporation

15. The component of an atom that contributes least to its mass is the

- (A) proton
- (B) nucleus
- (C) neutron
- (D) electron

16. An element will readily form an electrovalent compound if its electron configuration is

- (A) 2, 8, 1
- (B) 2, 8, 4
- (C) 2, 8, 8
- (D) 2, 8, 5

17. The most suitable metal that can be used as a lightening conductor is

- (A) silver
- (B) copper

- (C) iron
- (D) aluminum

18. The most abundant element on the earth's crust is

- (A) nitrogen
- (B) hydrogen
- (C) oxygen
- (D) fluorine

19. Metalloids are also referred to as

- (A) semi-metals
- (B) metals
- (C) colloids
- (D) non-metals

20. The ores that can be concentrated by floatation are

- (A) nitride ores
- (B) sulphide ores
- (C) oxide ores
- (D) chloride ores

BIOLOGY 2008/2009 ANSWERS

1. The s-orbital electrons move about to produce the effect of a spherical cloud round the nucleus. Thus s-orbitals are spherically symmetrical

- | | | | |
|------|-------|-------|-------|
| 2. A | 8. B | 14. B | 20. D |
| 3. A | 9. C | 15. D | |
| 4. D | 10. C | 16. A | |
| 5. C | 11. D | 17. D | |
| 6. A | 12. A | 18. A | |
| 7. C | 13. D | 19. A | |

**DELTA STATE UNIVERSITY, POST-UTME SCREENING PAST
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ENGLISH 2009/2010

Instruction: Choose the word opposite in meaning to the underlined word

1. The young engineer is good at **terminating** other people's products but has not been capable of ____ any of his own.

- (A) integrating
- (B) finishing
- (C) completing
- (D) initiating
- (E) organizing

2. The manager who expected to be given **respect** was treated with _____

- (A) dignity
- (B) scorn
- (C) shame
- (D) cruelty
- (E) disloyalty

3. Those who had invitation cards were **admitted** to the party while those who had none were _____

- (A) barred
- (B) repelled
- (C) expelled
- (D) compelled
- (E) restricted

4. Nobody expects him to show _____ for his children but he certainly bestows too much **affection** on them.

- (A) love
- (B) concern
- (C) intimacy
- (D) devotion
- (E) hatred

5. The challenger was **crude** and inexperienced in contrast to the champion who was _____

- (A) great
- (B) exposed
- (C) celebrated
- (D) refined
- (E) strong¹

From the words lettered A to E, choose the word that best complete each of the following sentences

6. The fishermen threw a stone into the river and this caused a _____

- (A) sprinkle
- (B) sparkle
- (C) splash
- (D) spring
- (E) storm

7. The play was so interesting that the _____ clapped for quite a long time at the end.

- (A) spectators
- (B) watchers
- (C) congregation
- (D) people
- (E) audience

8. The building _____ because of weak structural foundation

- (A) tumbled
- (B) succumbed
- (C) somersaulted
- (D) collapsed
- (E) caved

9. The magazine was _____ by the government for an offensive publication.

- (A) prescribed
- (B) proscribed
- (C) suspended
- (D) condemned
- (E) persecuted

10. Many people reacted to the brutal murder of the popular journalist with strong ____

- (A) indignation
- (B) demonstration
- (C) mobilization
- (D) condemnation
- (E) accusation

From the words of groups of words lettered A to E below each of the following sentences, choose the word or group of words that is nearest in meaning to the underlined word or group of words as used in the sentence

11. It takes a great deal of **stamina** to run the marathon race.

- (A) courage
- (B) determine
- (C) energy
- (D) intelligence
- (E) cleverness

12. But for the **principal** actor the play would have been dull.

- (A) important
- (B) head
- (C) master
- (D) famous
- (E) main

13. An open car has no protection against the **elements**

- (A) weather
- (B) emergency
- (C) molecule
- (D) atoms
- (E) atmosphere

14. He was **reluctant** to grant my request.

- (A) disposed
- (B) delighted
- (C) reticent
- (D) unwilling
- (E) agreeable

15. The detective was **perplexed** when the clues in the murder case pointed to at least a dozen different suspects
(A) surprised
(B) confused
(C) excited
(D) discouraged
(E) disappointed
16. The military governor called a **concerted** effort in solving the problems of the state.
(A) a dramatic
(B) an agitated
(C) a joint
(D) a directed
(E) an unfailing
17. My financial situation is so precarious that very soon I may be **insolvent**
(A) borrowing
(B) stealing
(C) soluble
(D) dependent
(E) bankrupt
18. The chairman is of the opinion that accepting the proposal would be **inimical** to the objectives of the association.
(A) harmful
(B) relevant
(C) irrelevant
(D) indispensable
(E) helpful
19. The famous politician was noted for his **pragmatic** approach to issues of national interest.
(A) idealistic
(B) romantic
(C) compromising
(D) practical
(E) inconsistent

20. Kunle is very **pessimistic** about our chance of success

- (A) sad
- (B) despondent
- (C) unconvinced
- (D) worried
- (E) concerned

ENGLISH 2009/2010 ANSWERS

- | | | | |
|------|-------|-------|-------|
| 1. D | 6. C | 11. C | 16. C |
| 2. B | 7. E | 12. E | 17. E |
| 3. A | 8. D | 13. A | 18. A |
| 4. E | 9. B | 14. D | 19. A |
| 5. D | 10. A | 15. B | 20. B |

CHEMISTRY 2009/2010

1. Which of the following is a physical change?

- (A) the bubbling of chlorine
- (B) the bubbling of chlorine into a jar containing hydrogen
- (C) the dissolution of sodium chloride in water
- (D) the passing of steam over heated iron

2. In the reaction $\text{SnO}_2 + 2\text{C} \rightarrow \text{Sn} + 2\text{CO}$, the mass of coke containing 80% carbon required to reduce 0.302kg of pure tin oxide is (Sn = 119, O=16 C=12)

- (A) 40kg
- (B) 0.20kg
- (C) 0.06kg
- (D) 0.0kg

3. The Avogadro number of 24g of magnesium is the same as that of

- (A) 1g of hydrogen molecules
- (B) 16g of oxygen molecules
- (C) 12g of carbon molecules
- (D) 35.5g of chlorine molecules.

4. If gas occupies a container of volume 146cm³ at 180C and 0.971atm, its volume in cm³ at STP is

- (A) 133
- (B) 146
- (C) 266

(D) 292

5. The volume occupied by 1.58g of a gas at STP is 500cm³. what is the relative molecular mass of the gas? (GMV at STP = 22.40dm³)

- (A) 28
- (B) 32
- (C) 44
- (D) 71

6. Equal volume of CO₂, SO₂, NO₂ and H₂S were released into a room at the same point and time. Which of the following gives the order of diffusion of the gases to the opposite corner of the room? (S=32, C=12; O=16, N=14, H=1)

- (A) CO, SO₂, NO₂, H₂S
- (B) SO₂, NO₂, SO₂, H₂S, CO
- (C) CO, H₂S, SO₂, NO₂
- (D) CO, H₂S, NO₂, SO₂

7. A basic postulate of the kinetic theory of gases is that the molecules of a gas move in straight lines between collisions. This implies that

- (A) collisions are perfectly elastic
- (B) forces of repulsion exist
- (C) forces of repulsion and attraction are in equilibrium
- (D) collisions are inelastic.

8. Which of the following terms indicates the number of bonds that can be formed by an atom?

- (A) oxidation number
- (B) valence
- (C) atomic number
- (D) electronegativity

9. $X(g) \rightarrow X(g)$: The type of energy involved in the above transformation is

- (A) ionization energy
- (B) Sublimation energy
- (C) lattice energy
- (D) electron affinity

10. Chlorine, consisting of two isotopes of mass number 35 and 37, has an atomic mass of 35.5. The relative abundance of the isotopes of the isotopes of mass number 37 is

- (A) 20
- (B) 25
- (C) 50
- (D) 75

11. 10.0dm³ of air containing H₂S as an impurity was passed through a solution of Pb(NO₃)₂ until all the H₂S had reacted. The precipitate of PbS was found to weigh 5.02g. According to the equation, $\text{Pb(NO}_3)_2 + \text{H}_2\text{S} \rightarrow \text{PbS} + 2\text{HNO}_3$ the percentage by volume of hydrogen sulphide (Pb = 207, S = 32 GMV at STP = 22.4dm³)

- (A) 50.2
- (B) 47.0
- (C) 4.70
- (D) 0.47

12. A blue solid, T, which weighed 5.0g was placed on a table. After 8 hours, the resulting pink solid was found to weigh 5.5g it can be inferred that substrate T. (A) is deliquescent

- (B) is hygroscopic
- (C) has some molecule of water of crystallization
- (D) is efflorescent.

13. The effluent of an industrial plant used plant used in the electrolysis of concentrated brine, with a flowing mercury cathode may contain impurities like

- (A) oxygen
- (B) hydrogen
- (C) mercury (II) chloride
- (D) hydrogen chloride

14. The solubility in moles per dm³ of 20g of CuSO₄ dissolved in 100g of water at 1800C is (Cu = 63.5; S = 32, O = 16)

- (A) 0.13
- (B) 0.25
- (C) 1.25
- (D) 2.00

15. Smoke consists of

- (A) solid particle dispersed in liquid
- (B) solid particles dispersed in gas
- (C) gas or liquid particles dispersed in liquid
- (D) liquid particles dispersed in liquid

16. $\text{Na}_2\text{C}_2\text{O}_4 + \text{CaCl}_2 \rightarrow \text{CaC}_2\text{O}_4 + 2\text{NaCl}$. Given a solution of 1.9g of sodium oxalate in 50g of water at room temperature, calculate the minimum volume of 0.1 M calcium chloride required to produce maximum calcium oxalate using the above equation

- (A) $1.40 \times 10^2 \text{ dm}^3$
- (B) $1.40 \times 10^{-2} \text{ dm}^3$
- (C) $1.40 \times 10^{-2} \text{ dm}^3$
- (D) $1.40 \times 10^{-2} \text{ dm}^3$

17. 2.0g of a monobasic acid was made up to 250cm³ with distilled water. 25.00cm³ of this solution required 20.00cm³ of 1.0M NaOH solution for complete neutralization. The molar mass of the acid is

- (A) 200g
- (B) 150g
- (C) 100g
- (D) 50g

18. What is the concentration of H⁺ ions in moles per dm³ of a solution of pH 4.398?

- (A) 4.0×10^{-5}
- (B) 0.4×10^{-5}
- (C) 4.0×10^{-3}
- (D) 0.40×10^{-3}

19. What volume of 11.0M hydrochloric acid must be diluted to obtain 1dm³ of 0.05M acid?

- (A) 4.5cm³
- (B) 5.65cm³
- (C) 6.76cm³
- (D) 7.78cm³

20. If 10.8g of silver is deposited in a silver coulometer connected in series with a copper coulometer, the volume of oxygen liberated is

(Ag = 108; Cu = 64; GMV at STP = 22.40dm³)

- (A) 0.56cm³
- (B) 5.60dm³
- (C) 11.20dm³
- (D) 22.40dm³

CHEMISTRY 2009/2010 SOLUTIONS/ANSWERS

1. C
2. STV
3. Avogadro's Number: at the same temperature and Pressure, equal volumes of all gases contain the same number of molecules. $N = 6.022 \times 10^{23}$ units of a specified item. $H_2 = 2.02$, $O_2 = 32.00$, $Cl_2 = 70.91$ = C
4. Using $P_1V_1 = P_2V_2$ thus $V_2 = 0.971 \times 146 \times 273$
 $T_1 T_2 1 \times 291 = 133 = A$
5. Using $PV = nRT$ but $n = m/M$ where m = given mass and M = Molar mass
 $PV = m RT$ and $M = mRT = 1.58 \times 8.314 \times 272$
 $M PV 101325 \times 5 \times 10^{-4} = 71 = D$
6. $CO_2 = 12 + 2(16) = 44$ $NO_2 = 14 + 2(16) = 46$
 $SO_2 = 32 + 2(16) = 64$ $H_2S = 2(1) + 32 = 34$
 The order should be: H_2S , CO_2 , NO_2 SO_2
7. = A
8. B
9. $X(g) \rightarrow X^+$
 (g) Z. Ionization energy
 $X(g) \rightarrow X^-$
 (g) Z. Electron affinity
10. $K \times 37 (100 - K) \times 35 = 35.5$
 100×100
 $0.37K + 35 - 0.35 = 35.5$
 $0.37K - 0.35K = 35.5 - 35$
 $0.02K = 0.5$ Thus $K \times 0.5 = 25 = B$
 0.02
11. $Pb(NO_3)_2 + H_2S \rightarrow PbS + 2HNO_3$
 1 mole of $H_2S \rightarrow 1$ mole of PbS
 $22.4dm^3 \rightarrow (207) + 32 = 239$
 $X \rightarrow 5.02$
 $X = 22.4dm^3 \times 5.02g = 0.47dm^3$

239g

Percentage by volume is $0.47\text{dm}^3 \times 100 = 4.7 = \text{C}$

10.0dm^3

12. = B

13. C

14. $\text{CuSO}_4 = 63.5 + 32 + 32 + 4(16) = 159.5$

Moles $= \frac{20}{159.5} = 0.125$

159.5

$100 \times 0.125 = 12.5$ and $1000 \times 0.125 = 1.25 = \text{C}$

100

15. = B

16. STV

17. STV

18. PH: The negative logarithm of the concentration (mol/L) of the H_3O^+ or (H^+)

ion; the commonly used scale ranges from 0 – 14.

$\text{pH} = -\log(\text{H}^+)$ and $-\log(\text{H}^+) = 4.398$

Multiplying through by -1 gives $\log(\text{H}^+) = -4.398$

Taking antilog of both sides gives $(\text{H}^+) = 10^{-4.398} = 4.0 \times 10^{-5}$

19. STV

20. If 10.8g of sulphur is deposited in silver coulometer connected in series with a copper coulometer, the volume of oxygen liberated is?

BIOLOGY 2009/2010

1. The nucleus is considered the central of a cell because it

(A) contains the genetic material

(B) contains the nuclear sap

(C) is bounded by the nuclear membrane

(D) is located at the centre of the cell

2. The prokaryotic cell type is characterized by

(A) complex cytoplasm in which different regions are poorly defined

(B) localization of different regions of the cell into tissue

(C) collection of organelles and macromolecular complexes

(D) simple cytoplasm with well defined regions

3. The natural tendency of organisms as they evolve is to

(A) decrease in size

(B) increase in number

- (C) develop specialized structures
- (D) feed indiscriminately

4. In snails, the hard calcareous shells are secreted by the

- (A) radula
- (B) tenidium
- (C) pneumostome
- (D) mantle

5. The ability of the cockroach to live in cracks and crevices is enhanced by the possession of

- (A) wings and segmented body
- (B) compound eyes
- (C) claws on the legs
- (D) dorso-ventrally flattened body

6. The caste of termite that lacks pigmentation is the

- (A) king
- (B) worker
- (C) soldier
- (D) queen

7. The structures that prevent food particles from escaping through the fish gills are called gill

- (A) arches
- (B) filaments
- (C) rakers
- (D) lamellae

8. A distinguishing feature of mammals is the possession of

- (A) skin
- (B) scale
- (C) nail
- (D) hair

9. Which of the following structures is capable of producing more tissues in the stem of a herbaceous flowering plant?

- (A) epidermis
- (B) pericycle

- (C) xylem
- (D) cambium

10. The manufacture of carbohydrates by plants takes place only in

- (A) the leaves
- (B) the green stems
- (C) chlorophyllous parts
- (D) flowering plants

11. In a water culture experiment, a plant showed poor growth and yellowing of the leaves. These may be due to deficiency of

- (A) copper
- (B) iron
- (C) magnesium
- (D) calcium

12. In millon's test, when the reagent is added to a protein food item, a white precipitate is produced which turns

- (A) blue on heating
- (B) yellow on heating
- (C) green on heating
- (D) red on heating

13. Regulation of sugar level takes place in the

- (A) pancreas
- (B) ileum
- (C) liver
- (D) kidney

14. Unicellular organisms transport essential nutrients direct to all parts of their bodies by the process of diffusion because they have

- (A) a large volume to surface area ratio
- (B) a large surface area to volume ratio
- (C) their bodies immersed in the nutrients
- (D) their outer membrane made of cellulose auricles

15. The heart of the adult frog consists of

- (A) two auricles and two ventricles
- (B) one auricle and one ventricle

- (C) two ventricles and one auricle
- (D) one ventricle and two auricles

16. In adult mammalian blood, the cells which lack nuclei are the

- (A) erythrocytes
- (B) lymphocytes
- (C) Leucocytes
- (D) phagocytes

17. Which of the following movements occurs during exhalation?

- (A) the diaphragm and intercostals muscles relax
- (B) the thoracic cavity increases in volume
- (C) the diaphragm and intercostals muscles contract
- (D) the diaphragm contracts and the intercostals muscles relax.

18. In which of the following groups of animals is the malpighian tubule found?

- (A) lizards, snakes and frogs
- (B) crickets, house flies and grasshoppers
- (C) millipedes, centipedes and scorpions
- (D) earthworms, roundworms and flatworms.

19. Which of the following is not a function of the mammalian skeleton?

- (A) protection
- (B) respiration
- (C) transportation
- (D) support

20. The most reliable estimate of growth is by measuring changes in

- (A) length
- (B) volume
- (C) surface area
- (D) dry weight

BIOLOGY 2009/2010 ANSWERS

- | | | | |
|------|-------|-------|-------|
| 1. A | 6. B | 11. B | 16. A |
| 2. A | 7. C | 12. D | 17. |
| 3. C | 8. D | 13. A | 18. B |
| 4. A | 9. A | 14. B | 19. C |
| 5. D | 10. C | 15. D | 20. D |

PHYSICS 2009/2010

1. Which of the following is not an example of a force?

- (A) tension
- (B) weight
- (C) friction
- (D) mass
- (E) thrust.

2. A body moves along a circular path with uniform angular speed of 0.6 rads^{-1} and at a constant speed of 3.0 ms^{-1} . Calculate the acceleration of the body towards the centre of the circle

- (A) 25.0 ms^{-1}
- (B) 5.4 ms^{-1}
- (C) 5.0 ms^{-1}
- (D) 1.8 ms^{-1}
- (E) 0.2 ms^{-1}

3. Which of the following is a derived unit?

- (A) Ampere
- (B) Kilogramme
- (C) second
- (D) Ohm
- (E) Kelvin

4. A boy timed 20 oscillations of a certain pendulum three times and obtained 44.3s, 45.5s and 45.7s respectively. Calculate the mean period of oscillation of the pendulum

- (A) 0.13s
- (B) 2.22s
- (C) 2.26s
- (D) 44.30s
- (E) 45.17s.

5. A particular starts from rest and moves with a constant accelerations of 0.5 ms^{-2} . Calculate the time taken by the particle to cover a distance of 25m

- (A) 2.5s
- (B) 7.1s
- (C) 10.0s
- (D) 50.0s

(E) 100.0s

6. A block of material of volume $2 \times 10^{-5} \text{m}^3$ is suspended from a spring balance with half the volume of the block immersed in water. What is the reading of the spring balance? (Density of water = $1.0 \times 10^3 \text{kg/m}^3$, $g = 10 \text{ms}^{-2}$)

(A) 0.10N

(B) 0.25N

(C) 0.30N

(D) 0.40N

(E) 0.50N

7. An object is projected with a velocity of 100ms^{-1} from the ground level at an angle to the vertical. If the total time of flight of the projectile is 10s, calculate ($g = 10 \text{ms}^{-2}$)

(A) 00

(B) 300

(C) 450

(D) 600

(E) 900

8. How far will a body move in 4 seconds if uniformly accelerated from rest at the rate of 2ms^{-2} ?

(A) 32m

(B) 24m

(C) 16m

(D) 12m

(E) 8m

9. If the temperature of water is gradually increased from 00 to 40C, the density of the water within this range

(A) increases for a while and then decreases

(B) decreases for a while and then increases

(C) increases gradually

(D) decreases gradually

(E) remains the same

10. The expansion of solid can be considered a disadvantage in the

(A) fire alarm system

(B) thermostat

- (C) riveting of steel plates
- (D) balance wheel of a watch
- (E) flitting of wheels on rims

11. A solid metal cube of side 10cm is heated from 100C to 600C. if the linear expansivity of the metal is $1.2 \times 10^{-5} \text{K}^{-1}$, Calculate the increase in its volume

- (A) 0.6cm³
- (B) 1.2cm³
- (C) 1.8cm³
- (D) 3.6cm³
- (E) 6.0cm³

12. A gas has a volume of 546cm³ at 00C. What is the volume of the gas at-1000C if its pressure remains constant?

- (A) 346cm³
- (B) 446cm³
- (C) 546cm³
- (D) 646cm³
- (E) 746cm³

13. An image which cannot be formed on a screen is said to be

- (A) inverted
- (B) real
- (C) virtual
- (D) erect
- (E) blurred

14. Longitudinal waves cannot be

- (A) diffracted
- (B) refracted
- (C) polarized
- (D) reflected
- (E) superposed

15. The images formed by a diverging lens are always

- (A) diminished, virtual and inverted
- (B) diminished, inverted and real
- (C) diminished, virtual and erect

- (D) magnified, virtual and erect
- (E) magnified, real and inverted.

16. In the normal use of a simple microscope, a person sees an

- (A) inverted, virtual and magnified image
- (B) erect, virtual and magnified image
- (C) erect, real and magnified image
- (D) inverted, real and magnified image
- (E) inverted and real image the same size as the object

17. A lens of focal length 15.0cm forms an upright image four times the size of an object, calculate the distance of the image from the lens

- (A) 11.ccm
- (B) 18.8cm
- (C) 37.5cm
- (D) 45.0cm
- (E) 75.0cm

18. An object is placed between two mirrors which are inclined at an angle of 120° and facing each other. Determine the number of images observed in two mirrors

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 5

19. In a ripple tank experiment, a vibrating plate is used to generate ripples in the water, if the distance between two successive troughs is 3.5cm and the wave travels a distance of 31.5cm in 1.5s, calculate the frequency of the vibrator

- (A) 3.0Hz
- (B) 6.0Hz
- (C) 12.0Hz
- (D) 27.0Hz
- (E) 73.5Hz

20. Which of the following have the longest wavelengths?

- (A) infra-red rays (B) gamma rays
- (C) x-rays
- (D) ultra-violet rays
- (E) radio waves)

PHYSICS 2009/2010 SOLUTIONS/ANSWERS

1. A FORCE is an agent that changes or tends to change the state of rest or of uniform motion in a straight line of a body but MASS is the quantity of matter or stuff contained in a body = D
2. $V = wr = 0.6 \times 3 = 1.8\text{m/s} = D$
3. = B
4. $44.3 + 45.5 + 45.7 = 45.17$ and $45.17 = 2.26\text{s} = C$
- 3 20
5. $u = 0, a = 0.5\text{ms}^{-2}, s = 25, t = ?$
 12
 using $v^2 = u^2 + 2as = 0 + 2(0.5)(25) = 25$
 if $v^2 = 25$ then $v = 5$
 now using $v = u + at$ we get $t = \frac{v - u}{a} = \frac{5 - 0}{0.5} = 10\text{s} = C$
 $a = 0.5$
6. From $T = 2U \sin \theta$, we see that $T_g = 2U \sin G$
 $= \sin^{-1}(T_g) = \sin^{-1} \left(\frac{10 \times 10}{2U} \right) = \sin^{-1}(0.5) = 30^\circ = B$
 $(2U) (2 \times 100)$
7. $S = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}(2)(42) = 16\text{m} = C$
8. = C
- 10 = C
11. $= 12 - 11 = e$
 $11(e = 11) (10) = 6.0\text{cm}^3 = E$
12. From Charles's Law $V_1 = V_2, V_2 = V_1 \frac{T_2}{T_1} = 546 \times \frac{173}{273} = 346\text{cm}^3 = A$
13. = C 14. C 15. C 16. B
17. $M = v = 4 \quad v = 4u$ where v – distance of image from lens and u –
 u distance of object from lens
 $v = -v = 4u$ (any upright image is virtual and thus negative)
 From $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ we put $\frac{1}{-4u} + \frac{1}{u} = \frac{1}{15}$ thus $\frac{3}{4u} = \frac{1}{15}$
 $4u = 45$ and $u = 11.25$. hence $v = 4u = 45.0\text{cm} = D$
18. = D
19. STV
20. C

MATHEMATICS 2009/2010

1. Simplify $125^{-1/3} \times 49^{-1/2} \times 100$

- (A) 350
- (B) 35
- (C) $1/35$
- (D) $1/350$
- (E) 0

2. If $32^x = 27$, what is x ?

- (A) 1
- (B) 1.5
- (C) 4.5
- (D) 18
- (E) 40.5

3. Express 0.00562 in standard form

- (A) 5.62×10^{-3}
- (B) 5.62×10^{-2}
- (C) 5.62×10^{-2x}
- (D) 5.62×10^2
- (E) 5.62×10^3

4. Given that $\frac{1}{3} \log_{10} P = 1$, find the value of P ,

- (A) $1/10$
- (B) 3
- (C) 10
- (D) 100
- (E) 1000

5. Simplify: $\log_8 (\log)$

- (A) $1/3$
- (B) $\frac{1}{2}$
- (C) $\frac{1}{3} \log 2$
- (D) $\frac{1}{2} \log 2$

6. If $\log x = 2.3675$ and $\log y = 0.9750$, find $x + y$, correct to three significant figures?

- (A) 1.18

- (B) 1.3
- (C) 9.03
- (D) 9.44
- (E) 9.46

7. While doing his Physics practical, Idowu recorded as 1.12cn instead of 1.21cm calculate his percentage error

- (A) 1.17%
- (B) 6.38%
- (C) 7.44%
- (D) 8.05%
- (E) 9.00%

8. Find the 4th term of an A.P whose first term is 2 and the common difference is 0.5

- (A) 0.5
- (B) 2.5
- (C) 3.5
- (D) 4
- (E) 4.5

9. An arc of length 22cm subtends an angle of θ at the centre of the circle. What is the value of θ if the radius of the circle is 15cm? (Taken $\pi = 22/7$)

- (A) 700
- (B) 840
- (C) 960
- (D) 1560
- (E) 1680

10. Find the sum of the first five terms of the GP 2,6,18,

- (A) 484
- (B) 243
- (C) 242
- (D) 130
- (E) 121

11. Let J be the set of positive integers. If $H = \{x: x \in J, x^2 < 3 \text{ and } x > 0\}$ then

- (A) $H = \{1\}$
- (B) H is an infinite set

- (C) $H = \{0, 1, 2\}$
- (D) $H = \{\}$
- (E) $J < H$

12. In a class of 80 students, every student has to study Economics or Geography, or both Economics and Geography if 65 students studied Economics and 50 studied Geography, how many studied both subjects?

- (A) 15
- (B) 30
- (C) 35
- (D) 45
- (E) 50

13. Factorize $x^2 + 4x - 192$

- (A) $(x-4)(x+48)$
- (B) $(x+48)(x+4)$
- (C) $(x-12)(x-16)$
- (D) $(x-12)(x-16)$
- (E) $(x+12)(x+16)$

14. Factorize $2e^2 - 3e + 1$

- (A) $(2e-1)(e-1)$
- (B) $(e+1)(2e+1)$
- (C) $(2e+3)(3-2)$
- (D) $(2e-3)(e-1)$
- (E) $(e^2-3)(2e-1)$

15. Solve the equation $7y^2 = 3y$

- (A) $y=3$ or 7
- (B) $y=0$ or 7
- (C) $y=0$ or $3/7$
- (D) $y=0$ or 9
- (E) $y=0$ or 10

16. Solve the equation $2a^2 - 3a - 27 = 0$

- (A) $3/2, 9$
- (B) $-2/3, 9$
- (C) $3, -9/2$
- (E) $-3, 9/2$

17. A sector of a circle of radius 7cm has an area of 44π cm². Find the angle of the sector.

- (A) 60
- (B) 260
- (C) 520
- (D) 1030
- (E) 2060

18. If the shadow of a pole 7m high is $\frac{1}{2}$ its length, what is the angle of elevation of the sun, correct to the nearest degree?

- (A) 90°
- (B) 63°
- (C) 60°
- (D) 26°
- (E) 0°

19. From the top of a building 10m high, the angle of depression of a stone lying on the horizontal ground is 69° . Calculate, correct to 1 decimal place the distance of the stone from the foot of the building

- (A) 3.6m
- (B) 3.8m
- (C) 6.0m
- (D) 9.3m
- (E) 26.1m

20. The bearing of a point X from Y is 074° . What is the bearing of Y from X?

- (A) 106°
- (B) 148°
- (C) 164°
- (D) 254°
- (E) 286°

MATHEMATICS 2009/2010 SOLUTIONS/ANSWER

1. $125 - \frac{1}{3} \times 49 - \frac{1}{2} \times 100 = 1\frac{1}{3} \times 1\frac{1}{2} \times 100 = 1 \times 1 \times 1 = 1$
 $125 - 49 - 50 = 26$ C
2. $32x = 27$ and $27 = 33$
 $2x = 3$ and $x = \frac{3}{2} = 1.5$ B
3. $5.62 \times 10^{-3} = A$
4. $\frac{1}{3} \log_{10} P = 1$ thus $\log_{10} P = 3$ B

**DELTA STATE UNIVERSITY, POST-UTME SCREENING PAST
QUESTIONS AND SOLUTIONS**

**ANSWER ALL QUESTIONS RELEVANT TO YOU:
SHADE THE ANSWER SHEET AS APPROPRIATE WITH HB PENCIL
ONLY**

ECONMICS 2010/2011

1. Economics is founded on the principle that:
 - a. Human wants are more than human needs
 - b. materials things are more than human wants
 - c. human want are limitless while the means to satisfy them are limited
 - d. human needs are easy to define

2. The capitalist economic system is characterized by all the following except:
 - a. Private ownership of the means of production
 - b. equitable distribution of income and wealth
 - c. Capital accumulation
 - d. markets and prices coordinate economic activities

3. If the equation relating consumption (c) to income (Y) is $C = N7,500 + 0.2Y$ then
 - a. Consumption is inversely related to income
 - b. consumption is the independent variable and income is the dependent variable
 - c. if income is N15,000, the consumption is N10,500
 - d. if consumption is N30,000, then income is N10,000

4. The reason for the law of demand can best be explained in terms of
 - a. supply
 - b. complementary goods the rationing function of prices
 - c, consumer
 - d. diminishing marginal utility

5. The demand curve for the pure monopolist is
 - a. perfectly price elastic
 - b. perfectly price inelastic
 - c. down sloping
 - d. up sloping

6. Which of the following statement is correct? The Central Bank

- a. prints the country's currency
 - b. handles the collection of debts owed to the government
 - c. lands money to businessmen
 - d. issues the country's currency
7. Which of the following is not a basic assumption underlying the theory of consumer behaviour
- a. consumers prefer more to less
 - b. consumers are rational
 - c. consumers are willing to make choices
 - d. consumers make consistent choices
8. The slope of the budget constraint
- a. changes as the marginal rate of satisfaction changes
 - b. is the ratio of the prices of the two goods
 - c. is the ratio of the budget of total utility
 - d. equals one, since the consumer can buy combination along the budget constraint
9. Disposable income equals
- a. personal income less business profit
 - b. personal income less taxes and subsidies
 - c. personal income less taxes plus subsidies
 - d. National income plus transfer payment
10. The 'term of trade' means
- a. the trade agreement between two countries
 - b. the difference in the volumes of export of two countries
 - c. the value of a unit of export in relation to the value of a unit of import
 - d. none of the above
11. The burden of tax on a commodity whose demand is infinitely inelastic
- a. is zero
 - b. will be borne by seller alone
 - c. will be borne by buyers alone
 - d. is impossible to tell
12. Inflation can be curbed by

- a. increasing aggregate demand
- b. paying higher wages
- c. reducing aggregate demand
- d. a deficit budget

13. The supply and demand for goods and services are influenced by the

- a. Supermarkets
- b. Entrepreneurs
- c. Mechanism of the law
- d. Price mechanism

14. In a sole proprietorship, the decisions are made by the

- a. Government
- b. Board of Directors
- c. Shareholder's conference
- d. Owner

15. Which of the following are direct taxes?

- A. income and company taxes
- b. sales taxes c. commodity taxes
- d. Excise duty

16. The production within the domestic territory of a country is called the

- a. Net National Product
- b. Gross Domestic Product
- c. Disposable Income
- d. Gross National Product

17. A producer sustains a loss in the short run if

- a. marginal revenue is less than price
- b. price is less than average cost
- c. marginal cost is less than marginal revenue
- d. average variable cost is less than average cost

18. Money market differs from capital market in that it:

- a. deals with short-term loans while capital market deals with long term funds
- b. deals with money only while capital market deals with capital as well
- c. is limited in scope while capital market is not
- d. uses interest rates while capital market does not

19. A production possibility curve shows:
- a. how much of the resources of society are used to produce a particular commodity
 - b. the rate of inflation
 - c. the rate of unemployment in the economy
 - d. the various combinations of two commodities that can be produced.
20. The situation in which the total amount of government spending exceeds total revenues is called a:
- a. balanced budget
 - b. budget deficit
 - c. budget surplus
 - d. fiscal surplus
21. The maximization of profit tends to be the driving force in the economic decision making of:
- a. entrepreneurs
 - b. workers
 - c. consumers
 - d. legislators
22. Which of the following equation is the correct one about Net National Product (NNP)?
- a. $NNP = \text{Gross National Product (GNP)} - \text{Depreciation}$
 - b. $NNP = \text{Gross National Product (GNP)} + \text{Net incomes from abroad}$
 - c. $NNP = \text{Gross Domestic Product (GDP)} - \text{Depreciation}$
 - d. $NNP = \text{Gross Domestic Product (GDP)} - \text{Net incomes from abroad}$
23. Middlemen in an economy perform the function of
- a. protection
 - b. hoarding
 - c. distribution
 - d. exchange
24. Line E in the diagram below represents:
- a. total cost
 - b. variable cost

- c. average cost
- d. fixed cost

25. Which of the following is the most important reason why wages differ among occupations?

- a. Length and cost of training
- b. Supply of Labour
- c. Attractiveness of the job
- d. Influence of trade Unions

ECONOMICS 2010/2011 ANSWER

- | | | | | |
|------|-------|-------|-------|-------|
| 1. C | 6. D | 11. C | 16. B | 21. A |
| 2. B | 7. A | 12. C | 17. B | 22. A |
| 3. C | 8. B | 13. D | 18. A | 23. C |
| 4. C | 9. C | 14. D | 19. D | 24. D |
| 5. A | 10. C | 15. A | 20. B | 25. A |

MATHEMATICS 2010/2011

1. If M represents the median and D the mode of the measurements 5, 9, 3, 5, 8 then (M,D) is

- A. (6,5)
- B. (5,8)
- C. (5,7)
- D. (5,5)
- E. (7,5)

2. A construction company is owned by two partners X and Y and it is agreed that their profit will be divided in the ratio 4:5. at the end of the year. Y received #5,000 more than x. what is the total profit of the company for the year?

- A. #20,000.00
- B. #25,000.00
- C. #30,000.00
- D. #15,000.003
- E. #45,000.00

3. Given a regular hexagon, calculate each interior angle of the hexagon.

- A. 600
- B. 300
- C. 1200
- D. 450
- E. 1350

4. Solve the following equations $4x - 3 = 3x + y = 2y + 5x - 12$

- A. $4x = 5, y = 2$
- B. $x = 2, y = 5$
- C. $x = -2, y = -5$
- D. $x = 5, y = -2$
- E. $x = -5, y = -2$

5. If $x = 1$ is root of the equation $x^3 - 2x^2 - 5x + 6$, find the other roots

- A. -3 and 2
- B. -2 and 2
- C. 3 and -2
- D. 1 and 3
- E. -3 and 1

6. If x is jointly proportional to the cube of y and the fourth power of z . In what ratio is x increased or decreased when y is halved and z is doubled?

- A. 4:1 increase
- B. 2:1 increase
- C. 1:4 decrease
- D. 1: 1 no change
- E. 3: 4 decrease

7 If $x + 2$ and $x - 1$ are factors of the expressions $lx + 2kx^2 + 24$, find the values of l and k

- A. $l = -6, k = -9$
- B. $l = -2, k = 1$
- C. $l = -2, k = -1$
- D. $l = 0, k = 1$
- E. $l = 6, k = 0$

8 The value of $(0.303)^3 - (0.02)^3$ is

- A. 0.019
- B. 0.0019

- C. 0.00019
- D. 0.000019
- E. 0.000035

9. The scores of a set of a final year students in the first semester examination in a paper are 41,29,55,21,47,70,70,40,43,56,73,23,50,50. Find the median of the scores.

- A. 47
- B. $48\frac{1}{2}$
- C. 50
- D. 48
- E. 49

10. If $26_3 + 441 = 714$, what number base has been used?

- A. 12
- B. 11
- C. 10
- D. 9
- E. 8

11. $0.00014323/1.940000 = k \times 10^n$ where $1 \leq k < 10$ and n is a whole number. The values of k and n are

- A. 7.381 and -11
- B. 2.34 and 10
- C. 3.87 and 2
- D. 7.831 and -11
- E. 5.41 and -2

12. P sold his bicycle to Q at a profit of 10%. Q sold it to R for ₦209 at a loss of 5%. How much did the bicycle cost P?

- A. ₦200
- B. ₦196
- C. ₦180
- D. ₦205
- E. ₦150

13. If the price of oranges was raised by $\frac{1}{2}k$ per orange, the number of oranges customer can buy for ₦2.40 will be less by 16. What is the present price of an orange?

- A. $2\frac{1}{2}k$
- B. $3\frac{1}{2}k$
- C. $5\frac{1}{2}k$
- D. $4\frac{1}{2}k$
- E. $6\frac{1}{2}k$

14. A man invested a total of #50,000 in two companies. If these companies pay dividend of 6% and 8% respectively, how much did he invest at 8% if the total yield is #3,700?

- A. #15,000
- B. #29,600
- C. #21,400
- D. #27,800
- E. #35,000

15. Thirty boys and x girls sat for a test. The mean of the boys' scores and that of the girls were respectively 6 and 8. Find x if the total score was 468.

- A. 38
- B. 24
- C. 36
- D. 22
- E. 41

16. The cost of production of an article is made up as follows

Labour - #70

Power - #15

Materials - #30

Miscellaneous - #5

Find the angle of the sector representing labour in a pie chart.

- A. 210°
- B. 105°
- C. 175°
- D. 150°
- E. 90°

17. Bola chooses at random a number between 1 and 300. What is the probability that the number is divisible by 4?

- A. $\frac{1}{3}$
- B. $\frac{1}{4}$

- C. $\frac{1}{5}$
- D. $\frac{4}{300}$
- E. $\frac{1}{300}$

18. A father leaves a legacy of N45 million for his children — Peter, David and Paul — to be shared in the ratio 7:5:3. What amount in million Naira would each receive respectively?

- A. N14, N7, N3
- B. N15, N5, N3
- C. N21, N15, N9
- D. N20, N16, N10

19. The expression $2\cos O + \sin^2 O$ has the numerical value

- A. 1
- B. 2
- C. 4
- D. 0

20. A regular polygon has each of its angles as 160° . What is the number of sides of the polygon?

- A. 36
- B. 9
- C. 18
- D. 20

21. Convert 241 in base 5 to base 8

- A. 718
- B. 1078
- C. 1768
- D. 2418

22. Find the least length of a rod which can be cut into exactly equal strips, each of either 40cm or 48cm in length.

- A. 120cm
- B. 240ccm
- C. 360cm
- D. 480cm

3. A rectangular has lawn has an area of 1815square yards. If its length is 50meters, find its width in metres. Given that 1meters equals 1.1yards

- A. 39.93
- B. 35.00
- C. 33.00
- D. 30.00

4. Reduce each number to two significant figures and then evaluate

$$\frac{(0.02174 \times 1.2047)}{0.023789}$$

- A. 0.8
- B. 0.9
- C. 1.1
- D. 1.2

5. A train moves from P to Q at an average speed of 90km/ hr and immediately returns from O to P through the same route and at an average speed of 45km/h. find the average speed for the centre journey.

- A. 55 00km/hr
- B. 60 00km/hr
- C. 67.50km/hr
- D. 75 00km/hr

MATHEMATICS 2010/2011 ANSWER

- | | | | | |
|------|------|------|------|-------|
| 1. D | 2. E | 3. C | 4. A | 5. C |
| 6. B | 7. D | 8. D | 9. B | 10.D |
| 11.A | 12.A | 13.A | 14.E | 15.C |
| 16.A | 17.C | 18.C | 19.A | 20. C |
| 21.B | 22.B | 23.C | 24.C | 25.C |

ENGLISH 2010/2011

Comprehension

Read the following passages carefully and answer the questions that follow.

Passage 1

The best acceptable definition of history is that it is a record of the past actions of mankind, based on surviving evidence. It is this evidence that the historian employs to chronicle and correlate events, by which he arrives at conclusions which he believes to be valid. Hence, the historian is referred to as an interpreter of the development of mankind. It should be understood that there is more than one way of treating the past.

For example, in trying to deal with the revolutions in Nigeria, past and present, the historian may describe the events in a narrative order. Or, he may choose to concentrate on analysis of the general causes, comparing their stages of evolution with the patterns of revolution in other countries. The historian does not seek to attain the same kind of results as the scientist, who can verify his conclusions by repeating his experiment under controlled conditions. Whilst he also attempt to classify the phenomena, the historian is more likely to consider event in terms of their uniqueness. Added to this is the fact that history is concerned, fundamentally, with the lives and actions of men, and as such, the historian's search for causes is bound to be relatively subjective, as compared to that by the scientist.

In essence, however, historians are agreed and insist that history should be written as scientifically as possible and that the evidence should be analyzed with the same objective attitude employed by the scientist when he examines certain phenomena of nature.

1. According to the passage, a historian should try to examine a material
 - A. scientifically
 - B. subjectively
 - C. accurately
 - D. objectively
2. According to the passage, one of the duties of a historian is
 - A. to predict the future
 - B. to analyse the past and future
 - C. to explain the significance of past events

- D. to interpret the development of mankind
3. The scientist tends to be more reliable than historian because
- A. he works in a laboratory
 - B. he is better qualified
 - C. he can crosschecked his result several time
 - D. he has more time to work at his Experiments
4. Flow can history be scientifically recorded?
- A. by examining available evidence and analyzing unusual occurrences
 - B. by falsifying and fabricating available facts
 - C. by speculating on what was and ought to have been
 - D. a record of the present actions of mankind based on surviving evidence
 - E. a record of the past action of mankind based on surviving evidence

Passage 2

From the apex of the Niger Delta southwards, dry land, overgrown with dense forests still virginal in various spots, gives way to seasonally inundated zones. Here, sweet water swamps with strands of raffia palms gradually merge into tidal swamps of brackish ooze, where mud skippers thrive under the arching roots of mangroves. The Niger, fingering through a thousand creeks, meet the sea in a dozen estuaries. Strong rives currents drifts and mud across the river mouths, scaling tern again and again to navigation.

6. Where do mudskippers thrive?
- A. in the creeks
 - B. in the swamps
 - C. in the mangroves
 - D. in the roots
7. "Fingering through" as used in the passage means
- A. cutting across
 - B. passing through
 - C. cutting between
 - D. passing between
8. According to the passage, how would you describe a seasonally inundated zone?
- A. a zone always covered with mud

- B. a zone always covered with shallow water
 - C. a zone under water at certain times of the year
 - D. a zone subject to heavy rain every season
9. Where does the Niger meet the sea?
- A. in the creeks
 - B. in the Delta
 - C. in the swamps
 - D. in the forest
10. What is brackish ooze?
- A. a strong river current
 - B. a mixture of fresh water and mud
 - C. a mixture of fresh water and salt water
 - D. fresh and clean water

From the list of words lettered A - E choose the word that is most nearly opposite in meaning to the word underlined and which will appropriately fill the gap in the sentence. (11 - 15)

11. Athletes wishing to get rid of their and get more energy should take more exercise
- A. fat
 - B. oxygen
 - C. lethargy
 - D. trainers
 - E. despots
12. A metal will expand when it's heated and when it cools
- A. shorten
 - B. lesser
 - C. contract
 - D. congeal
 - E. curtail
13. The political aspirant asked the villagers to support him and not to his authority in any way
- A. deny
 - B. undermine

- C. defy
- D. despite
- E. attack

14. She was a very proficient hairdresser but had little aptitude for sewing in which she was

- A. new
- B. unskilled
- C. unlearned
- D. ignorant
- E. awkward

15. If you do not accept the offer of a job in the Secretariat within the next one week, we shall assume you it

- A. denied
- B. refused
- C. deprived
- D. left
- E. lost

From the list of words lettered A - E below each of the sentences, choose the one which is nearest in meaning to the underline word as it is used in the sentences (16 - 20)

16. The examiner said that the candidate's performance in the examination was not good enough

- A. failure
- B. achievement
- C. success
- D. presentation
- E. marks

17. After finishing the 800 meters race, he fell asleep from exhaustion.

- A. weakness
- B. fatigue
- C. overwork
- D. eagerness
- E. sloth

18. Last night there was a very fierce rain Storm
A. raging
B. storming
C. angry
D. violent
E. ferocious.
19. After warn, on our way to Benin, we passed through a dense forest
A. crowded
B. close
C. thick
D. heavy
E. wooded
20. The footballers went back to their camp sullenly
A. cheekily
B. quickly
C. stubbornly
D. resentfully
E. silently

In each of the following questions 21- 25, fill each gap with the appropriate option from the list. Following exercises express different times by using different tenses.

From the natives suggested, choose any one that best suits each context

21. More..... to your elbow as you campaign for press freedom.
A. energy
B. power
C. effort
D. grease
E. kinetic
22. Sir, I'm not lying about the matter, I know nothing of it, if I knew
A. I must tell you
B. I can tell you
C. I would tell you
D. I shall tell you
E. I will tell you

23. The editor was not happy that the Nigeria press was hemmed
- A. up
 - B. across
 - C. in
 - D. over
 - E. sideways
24. A child that shows mature characteristics at an early age may be described as
- A. precocious
 - B. ingenuous
 - C. premature
 - D. preconceived
25. That is a very terrible woman; everyday she makes a lot of noise about one thing or the other. I'm not surprised, that's what her sisterstoo.
- A. are used to doing
 - B. do
 - C. always used to do
 - D. are doing

ENGLISH 2010/2011 ANSWER

1.D	2.D	3.C	4.A	5.D
6.C	7.B	8.D	9.B	10.C
11.C	12.C	13.B	14.B	15.B
16.B	17.B	18.A	19.C	20.D
21.D	22.C	23.C	24.A	25.B

GOVERNMENT 2010/2011

1. Who formed the first political party in Nigeria?
 - A. obasanjo
 - B. osama
 - C. obi
 - D. Herbert Macaulay

2. What was the first political party in Nigeria?
 - A. APC
 - B. PDP
 - C. NNDP
 - D. AD

3. Who is the current president of Nigeria?
 - A. Obasanjo
 - B. Saraki
 - C. Buhari
 - D. Jonathan

4. Who is the current vice president of Nigeria?
 - A. Osinbanja
 - B. Atiku
 - C. Peter Obi
 - D. Namadi Sambo

5. What does the eagle in the Nigerian coat of arm represent?
 - A. peace
 - B. Terror
 - C. Agriculture
 - D. Strength

6. Representative democracy is best characteristic by
 - A. free elections and proper registers of voters
 - B. a politically educated electorate
 - C. rule by the interest group
 - D. proper constituencies and a real choice of candidates

7. While political parties aim at forming a government pressure groups aim at

- A. causing social unrest
 - B. influencing governmental decisions
 - C. controlling nation's economy
 - D. getting workers to unite
8. When the electorate vote for representatives who is turn vote on their behalf we say it is
- A. an indirect election
 - B. an unfair election
 - C. a disputed election
 - D. a rigged electron
9. An election which is conducted to fill a vacant seat in a legislature is called a
- A. by election
 - B. general election
 - C. referendum
 - D. plebiscite
10. Which of these countries does NOT operate a federal constitution?
- A. USA
 - B. Nigeria
 - C. Canada
 - D. France
11. The major advantage of the secret ballot is that
- A. it is faster than other systems
 - B. nobody can be prevented from voting
 - C. it ensures the anonymity of each voter
 - D. losers can ask for another secret voter
12. In a one party state
- A. there are no free citizens
 - B. the communist party is the only legal party
 - C. the ruling party is the only legal
 - D. elections to be legislature are held at the party's conferences
13. A cabinet system of government is practiced in
- E. Britain and Canada
 - F. the soviet union

- G. all European Countries including Britain
 - H. the United State of America
14. A proclamation by the head of state ending a session of parliament is called
- I. a dissolution
 - J. an adjournment
 - K. a prorogation
 - L. a devolution
15. The constitution of the federal republic of Nigeria
- A. promotes unity of diversity
 - B. allows for the dominance of the minority ethnic groups
 - C. concentrates governmental power at on one level of government
 - D. ensures the dominance of one political party.
16. The transfer of authority to local government council is known as
- M. delegation
 - N. fusion
 - O. fragmentation
 - P. devolution
17. The principle of anonymity of civil servants means that they
- A. are not the servant of a particular government
 - B. are trained for the duties they perform
 - C. are credited or blamed for anything they do
 - D. have a career
18. The six registered political parties in Nigeria in 1982 were
- a. UPN, NPN, PPA, PPP, NPP, and NAP
 - b. UPN, GNPP, NAP, PRP, NPP, NPN,
 - c. NPN, UPN, NPP, PRP, PPP NNDP,
 - d. NPC, GNPP, PRP, UPN, NPP and PPA
19. Which of the following is NOT a pressure group in Nigeria
- A. farmers
 - B. the Nigerian union of teachers (NUT)
 - C. Nigeria medical associate (NMA)
 - D. academic staff union of university (ASUU)
20. The N.C.N.C sent a delegation to London in 1947 to protest against the

- A. burns constitution
 - B. Richards constitution
 - C. Littleton constitution
 - D. Macpherson constitution
21. A popular principle of colonial administration in British west Africa was
- A. association
 - B. indirect rule
 - C. paternalism
 - D. assimilation
22. The first governor-general of colonial Nigeria was
- A. sir, Hugh Clifford
 - B. sir James Robertson
 - C. lord Lugard
 - D. sir Ralph moore.
23. The first military government in Nigeria was headed by
- A. General Yakubu Gowon
 - B. General Agunyi Ironsi
 - C. General M. Mohammed
 - D. General O. Obasanjo.
24. The first general election in Nigeria was hold in
- A. 1933
 - B. 1952
 - C. 1955
 - D. 1959
25. The supreme organ of the U.N O is the
- A. general assembly
 - B. secretary general
 - C. world court
 - D. world bank

GOVERNMENT 2010/2011 ANSWER

1.D 2.C 3.C 4.A 5.D 6.A 7.B 8.A 9.A 10D
11C 12C 13A 14A 15A 16A 17C 18B 19A 20B
21B 22C 23B 24D 25A

**DELTA STATE UNIVERSITY, POST-UTME SCREENING PAST
QUESTIONS AND SOLUTIONS**

ANSWER ALL QUESTION:

**SHADE THE ANSWER SHEET AS APPROPRIATE WITH HB PENCIL
ONLY**

USE OF ENGLISH 2012/2013

Choose the correct option to fill the blank spaces

1. The guests breakfast by the time the bus arrives.
 - a. shall have finished
 - b. have finished
 - c. had finished
 - d. Are finishing

2. You can go on, I what you are saying
 - a. will be understanding
 - b. an understanding
 - c. understand
 - d. Was understanding

3. When he was knocked on the head, he fell to the ground
 - a. fainted
 - b. unconscious
 - c. Collapsed
 - d. noisily

4. It so hard that all the cars have stopped moving.
 - a. will rain
 - b. rained
 - c. Rain
 - d. Rains

5. The principal will be going away on leave, in his absence, the vice-.principal will....the school
 - a. overlook
 - b. take over
 - c. Look after
 - d. care for

6. In 1966, the military... because of the tragic failure of the civilian politicians

- a. took to power
- b. undertook power
- c. came to power
- d. Marched into power

7. The way to stop some frivolous publications is to the press

- a. shackle
- b. gag
- c. Bind
- d. Feller

8. A good film, wasn't it?

- a. it is
- b. there is
- c. It was
- d. Was it

9. It was a blind alley. I would not walk any

- a. further
- b. farther
- c. Far
- d. Furthest

Read the passage below and answer questions 10-14 choosing the correct option

There is one fascinating question that arises out of the contemplation of mud sculpture: why should anybody use unbaked mud, the most perishable of materials. Is it because no other materials are readily available? Mud question is not easy to answer definitely. Mud is of course the cheapest and most readily available materials. Yet there ample proof that mud is not used merely because it is easy to get hold of and cheap. Many Igbo Mbari houses are the only buildings in the village that have an important corrugated iron roof which shows that the people who built them sun no cost to make them look important.

Materials are used for their mystic properties of absorbing or repelling human radiation while mud is considered the appropriate medium for Ala (the Igbo earth goddess) Olukun the Bini god of the ocean, the Obatala worshippers use ivory as a protection in the destructive psychic influence of a man whose mentality is different or opposed to his.

10. 'Mud sculpture' as used in the passage means
- mud houses
 - mud paintings
 - objects made of mud
 - Mud goddess
11. The Ibos often use corrugated iron for the roofs of their Mbari houses
- because it is expensive
 - because it lasts longer than thatch
 - in order to make them look important
 - because it is less imposing than thatch
12. Different materials are chosen because
- they are thought to have particular physical powers
 - they are associated with different deities
 - some are difficult to obtain
 - because they are used by different worshippers of the same god.
13. It seems probable that mud is used for sculpture primarily because
- it has the quality of durability
 - it is cheap
 - it is easy to obtain
 - it is thought to have a particular significance.
14. A suitable title for the passage is
- mud houses
 - mud sculpture
 - materials for sculpture
 - materials decoration of houses.

In each of the questions, choose the word opposite in meaning to the underlined word or phrase in italics

15. Many Nigerians are wary of night journeys.
- satisfied with
 - troubled about
 - worried about
 - Unscarred of
16. We cannot but talk about his invaluable contributions to the affairs of the

- a. offensive
- b. worthless
- c. Unrecognized
- d. incalculable

17. The hasty action will bring nothing but discredit to them

- a. honour
- b. shame
- c. Disgrace
- d. Reward

From the words lettered A to E choose the word nearest in meaning to the underlined word

18. One of the surest ways to ensure good health is to have a wholesome and adequate diet

- a. health giving
- b. mixed
- c. Palatable
- d. hygienic

19. All efforts made to settle the quarrel were futile

- a. successful
- b. wicked
- c. Fruitless
- d. remarkable

20. The manager often quarrels with his staff because he goes around with a chip on his shoulder

- a. reasons to be angered
- b. deformed c. look of contempt
- d. Search for gossip.

21. The corrupt official had to leave the public service willy nilly.

- a. reluctantly
- b. unprepared
- c. willingly
- d. Compulsorily

22. We will surely buy this idea of introducing entirely new product.

- a. pay for
- b. accept
- c. Trade in
- d. Barter

23. This year's harvest is surplus

- a. more than enough
- b. limitless
- c. Plenty
- d. less than enough

24. I would like you to do the work piecemeal

- a. at meal time
- b. bit by bit
- c. Badly
- d. as a masterpiece

25. The news of his performance in the session examination has made Toristeju crestfallen

- a. pompous
- b. elated
- c. Highly irritable
- d. happy.

FEDERAL UNIVERSITY OF AGRICULTURE, ABEOKUTA POST-UTME SCREENING PAST QUESTIONS AND SOLUTIONS

ANSWER ALL QUESTION:

SHADE THE ANSWER SHEET AS APPROPRIATE WITH HB PENCIL ONLY

GENERAL PAPER 2012/2013

1. The major raw material for pottery is.....

- A. Metal
- B. Ceramics
- C. Cement
- D. Clay

ANS D

2. Which of the following will sink when placed on water?

- A. ball
- B. plastic cup
- C. balloon
- D. eraser

ANS D

3. The following are domestic birds except

- A. duck
- B. pigeon
- C. turkey
- D. vulture

ANS D

4. The degree of hotness and coldness of a place is called

- A. rainfall
- B. humidity
- C. temperature
- D. weather

ANS C

5. A place where two rivers meet is called

- A. confluence
- B. junction
- C. joint

D. delta

ANS A

6. Which of the following is not a mammal

A. whale

B. dog

C. cat

D. shark

ANS D

7. How many Lunar months make one year

A. 13

B. 11

C. 12

D. 14

ANS A

8. Nigeria started using N200 notes on .

A. 1st Feb, 2000

B. 5th May, 2001

C. 20th August, 2000

D. 1st Nov. 2000.

ANS D

9. Queen Elizabeth visited Nigeria for the second time on .

A. 3rd Dec. 2003

B. 2nd October. 1980

C. 7th June 1985

D. 14th June 1979

ANS A .

10. The International organization responsible for taking care of the injured is known as

A. Red Cross Society

B. UNESCO

C. World Bank

D. Man O'War.

ANS A

11. Which of the following is linked with CONFLUENCE ·

- A. Niger State
- B. Kogi State
- C. Benue State
- D. None of above

ANS B

12. A patient suffering from Night blindness requires ·

- A. Vitamin A
- B. Vitamin E
- C. Calcium only
- D. Iodine

ANS A

13. Zebra crossing is for -----

- A. Playground
- B. Easy transport action of Zebra
- C. Safety of Pedestrians
- D. Zoological garden.

ANS C

14. In which part of Africa will you find Tanzania

- A. West Africa
- B. North Africa
- C. Southern Africa
- D. East Africa

ANS D

15. Today is Monday. After 63 days, it will be:

- A. Wednesday
- B. Monday
- C. Tuesday
- D. Saturday

ANS B

16. The title of the traditional ruler of Benin is called.....

- A. Obi
- B. Obong
- C. Oba

D. Igwe
ANS C

17. Which of these is not a method of preventing iron from rusting?

- A. Lubrication
- B. Galvanization
- C. Melting
- D. Painting

ANS C

18. Which of the following is not a rainbow colour?

- A. Indigo
- B. Purple
- C. Orange
- D. Green

ANS B

19. A person who flies an aircraft is called

- A. captain
- B. craftologist
- C. a pilot
- D. an aeronautics engineer

ANS C

20. Kwame Nkrumah Museum is located in

- A. Tanzania
- B. Kenya
- C. Mauritania
- D. Ghana

ANS D

21. An example of confluence town in Nigeria is

- A. Onitsha
- B. Obudu
- C. Lokoja
- D. Port Harcourt

ANS C

22. A trained person that specializes in the treatment of diseases and disorders of the teeth is called.....

- A. Cardiologist
- B. Dermatologist
- C. Physician
- D. Dentist

ANS D

23. In which country is Mississippi river

- A. USA
- B. Argentina
- C. Brazil
- D. Canada.

ANS A

24. In parliamentary system of government, members of the parliament are required to report the proceedings of the house to their.....

- A. Local government chairman and governors
- B. Constituencies
- C. Political parties and party stalwarts
- D. Traditional rulers

ANS B

25. One essential duty of a citizen to his/her state is to.....

- A. Vote and support the government in power
- B. Recite the national anthem and the pledge
- C. Pay his tax
- D. Encourage other citizens to be loyal

ANS C

26. Under whose regime were Delta and Adamawa States created?

- A. Gen Sanni Abacha
- B. Gen yakubu Gowon
- C. Gen Murtala Muhammed
- D. Gen Ibrahim Babangida

ANS D

27. The body charged with the responsibility of regulating and protecting consumers against illicit foods and drugs in Nigeria

- A. Consumer protection council
- B. Standard Organization of Nigeria
- C. National Drug Law Enforcement agency
- D. National Agency for Food, Drug Administration and Control

ANS D

28. The senate president during the second republic was.....

- A. Dr. Joseph Wayas
- B. Senator Anyim Pius Anyim
- C. Senator Iyorchia Ayu
- D. Chief Adopous Wabara

ANS A

29. We eat Carbohydrate food to give us.....

- A. Energy
- B. Good sight
- C. Good teeth
- D. Strong bones

ANS A

30. President Barrack Obama is the president of the United States of America and first African-American president of U.S.

- A. 35th
- B. 44th
- C. 40th
- D. 39th

ANS B

USE OF ENGLISH 2012/2013

Comprehension

Read the following passages carefully and answer the questions that follow.

Passage 1

The best acceptable definition of history is that it is a record of the past actions of mankind, based on surviving evidence. It is this evidence that the historian employs to chronicle and correlate events, by which he arrives at conclusions which he believes to be valid. Hence, the historian is referred to as an interpreter of the development of mankind. It should be understood that there is more than one way of treating the past.

For example, in trying to deal with the revolutions in Nigeria, past and present, the historian may describe the events in a narrative order. Or, he may choose to concentrate on analysis of the general causes, comparing their stages of evolution with the patterns of revolution in other countries. The historian does not seek to attain the same kind of results as the scientist, who can verify his conclusions by repeating his experiment under controlled conditions. Whilst he also attempt to classify the phenomena, the historian is more likely to consider event in terms of their uniqueness. Added to this is the fact that history is concerned, fundamentally, with the lives and actions of men, and as such, the historian's search for causes is bound to be relatively subjective, as compared to that by the scientist.

In essence, however, historians are agreed and insist that history should be written as scientifically as possible and that the evidence should be analyzed with the same objective attitude employed by the scientist when he examines certain phenomena of nature.

1. According to the passage, a historian should try to examine a material
 - E. scientifically
 - F. subjectively
 - G. accurately
 - H. objectively
2. According to the passage, one of the duties of a historian is
 - E. to predict the future
 - F. to analyse the past and future
 - G. to explain the significance of past events
 - H. to interpret the development of mankind
3. The scientist tends to be more reliable than historian because
 - E. he works in a laboratory
 - F. he is better qualified
 - G. he can crosschecked his result several time
 - H. he has more time to work at his Experiments
4. Flow can history be scientifically recorded?
 - F. by examining available evidence and analyzing unusual occurrences
 - G. by falsifying and fabricating available facts
 - H. by speculating on what was and ought to have been

- I. a record of the present actions of mankind based on surviving evidence
- J. a record of the past action of mankind based on surviving evidence

Passage 2

From the apex of the Niger Delta southwards, dry land, overgrown with dense forests still virginal in various spots, gives way to seasonally inundated zones. Here, sweet water swamps with strands of raffia palms gradually merge into tidal swamps of brackish ooze, where mud skippers thrive under the arching roots of mangroves. The Niger, fingering through a thousand creeks, meet the sea in a dozen estuaries. Strong rives currents drifts and mud across the river mouths, scaling tern again and again to navigation.

- 6. Where do mudskippers thrive?
 - E. in the creeks
 - F. in the swamps
 - G. in the mangroves
 - H. in the roots
- 7. "Fingering through" as used in the passage means
 - E. cutting across
 - F. passing through
 - G. cutting between
 - H. passing between
- 8. According to the passage, how would you describe a seasonally inundated zone?
 - E. a zone always covered with mud
 - F. a zone always covered with shallow water
 - G. a zone under water at certain times of the year
 - H. a zone subject to heavy rain every season
- 9. Where does the Niger meet the sea?
 - E. in the creeks
 - F. in the Delta
 - G. in the swamps
 - H. in the forest
- 10. What is brackish ooze?
 - E. a strong river current

- F. a mixture of fresh water and mud
- G. a mixture of fresh water and salt water
- H. fresh and clean water

From the list of words lettered A - E choose the word that is most nearly opposite in meaning to the word underlined and which will appropriately fill the gap in the sentence. (11 - 15)

11. Athletes wishing to get rid of their and get more energy should take more exercise

- F. fat
- G. oxygen
- H. lethargy
- I. trainers
- J. despots

12. A metal will expand when it's heated and when it cools

- F. shorten
- G. lesser
- H. contract
- I. congeal
- J. curtail

13. The political aspirant asked the villagers to support him and not to his authority in any way

- F. deny
- G. undermine
- H. defy
- I. despite
- J. attack

14. She was a very proficient hairdresser but had little aptitude for sewing in which she was

- F. new
- G. unskilled
- H. unlearned
- I. ignorant
- J. awkward

15. If you do not accept the offer of a job in the Secretariat within the next one week, we shall assume you it

- F. denied
- G. refused
- H. deprived
- I. left
- J. lost

From the list of words lettered A - E below each of the sentences, choose the one which is nearest in meaning to the underline word as it is used in the sentences (16 - 20)

16. The examiner said that the candidate's performance in the examination was not good enough

- F. failure
- G. achievement
- H. success
- I. presentation
- J. marks

17. After finishing the 800 meters race, he fell asleep from exhaustion.

- F. weakness
- G. fatigue
- H. overwork
- I. eagerness
- J. sloth

18. Last night there was a very fierce rain Storm

- F. raging
- G. storming
- H. angry
- I. violent
- J. ferocious.

19. After warn, on our way to Benin, we passed through a dense forest

- F. crowded
- G. close
- H. thick
- I. heavy

J. wooded

20. The footballers went back to their camp sullenly

- F. cheekily
- G. quickly
- H. stubbornly
- I. resentfully
- J. silently

In each of the following questions 21- 25, fill each gap with the appropriate option from the list. Following exercises express different times by using different tenses.

From the natives suggested, choose any one that best suits each context

21. More..... to your elbow as you campaign for press freedom.

- F. energy
- G. power
- H. effort
- I. grease
- J. kinetic

22. Sir, I'm not lying about the matter, I know nothing of it, if I knew

- F. I must tell you
- G. I can tell you
- H. I would tell you
- I. I shall tell you
- J. I will tell you

23. The editor was not happy that the Nigeria press was hemmed

- F. up
- G. across
- H. in
- I. over
- J. sideways

24. A child that shows mature characteristics at an early age may be described as

- E. precocious

- F. ingenuous
- G. premature
- H. preconceived

25. That is a very terrible woman; everyday she makes a lot of noise about one thing or the other. I'm not surprised, that's what her sisterstoo.

- E. are used to doing
- F. do
- G. always used to do
- H. are doing

ENGLISH 2012/2013 ANSWERS

1.D	2.D	3.C	4.A	5.D
6.C	7.B	8.D	9.B	10.C
11.C	12.C	13.B	14.B	15B
16.B	17.B	18.A	19.C	20.D
21.D	22.C	23.C	24.A	25.B

MATHEMATICS 2012/2013

1. Convert 241 in base 5 to base 8

- A. 718
- B. 1078
- C. 1768
- D. 2418

2. Find the least length of a rod which can be cut into exactly equal strips, each of either 40cm or 48cm in length.

- A. 120cm
- B. 240ccm
- C. 360cm
- D. 480cm

3. A rectangular has lawn has an area of 1815square yards. If its length is 50meters, find its width in metres. Given that 1meters equals 1.1yards

- A. 39.93
- B. 35.00
- C. 33.00
- D. 30.00

4. Reduce each number to two significant figures and then evaluate $(0.02174 \times 1.2047) \div 0.023789$
- A. 0.8
 - B. 0.9
 - C. 1.1
 - D. 1.2
5. A train moves from P to Q at an average speed of 90km/hr and immediately returns from Q to P through the same route and at an average speed of 45km/h. find the average speed for the entire journey.
- A. 55 00km/hr
 - B. 60 00km/hr
 - C. 67.50km/hr
 - D. 75 00km/hr
6. If the length of a square is increased by 20% while its width is decreased by 20% to form a rectangle, what is the ratio of the area of the rectangle to the area of the square?
- A. 6.5
 - B. 25.24
 - C. 5.6
 - D. 24.25
7. Two brothers invested a total of ₦5,000.00 on a farm project. The farm yield was sold for ₦ 15, 000.00 at the end of the season. If the profit was shared in the ratio 2:3, what is the difference in the amount of profit received by the brothers?
- A. ₦2,000.00
 - B. ₦4,000.00
 - C. ₦6,000.00
 - D. ₦10,000.00
8. Peter's weekly wages are ₦20.00 for the first 20 weeks and ₦36.00 for the next 24 weeks. Find his average weekly wage for the remaining 8 weeks of the year. If his average weekly wage for the whole year is ₦30.00
- A. ₦37.00
 - B. ₦35.00
 - C. ₦30.00

- D. #5.00
9. A man invests a sum of money at 4% per annum simple interest. After 3 years, the principal amounts to #7,000.00. find the sum invested
- A. #7,840.00
 - B. #6,250.00
 - C. #6,160.00
 - D. #5,833.33
10. By selling 20 oranges for #1.35 a trader makes a profit 8%. What is his percentage gain or loss if he sells the same 20 oranges for #1.10?
- A. 8%
 - B. 10%
 - C. 12%
 - D. 15%
11. Four boys and ten girls can cut a field in 5 hours. If the boys work at $\frac{1}{4}$ the rate of which the girls work, how many boys will be needed to cut the field in 3 hours?
- A. 180
 - B. 60
 - C. 25
 - D. 20
12. Evaluate without using tables.
- A. $\frac{625}{8}$
 - B. $\frac{8}{625}$
 - C. $\frac{1}{8}$
 - D. 8
13. Instead of writing $\frac{35}{6}$ as a decimal correct to 3 significant figures, a student wrote it correct to 3 places of decimals. Find his error in standard form
- A. 0.003
 - B. 3.0×10^{-3}
 - C. 0.3×10^2
 - D. 0.3×10^{-3}
14. Simplify without using tables $(\log 26 - \log 23) / (\log 28 - 2\log 21/2)$
- A. $\frac{1}{5}$

- B. $\frac{1}{2}$
- C. $-\frac{1}{2}$
- D. $\log_{23}/\log_{27}$

MATHEMATICS 2012/2013 ANSWER

1B 2B 3C 4C 5C 6 D 7A 8A 9B 10C 11D 12A 13B

BIOLOGY 2012/2013

1. One of the functions of xylem
 - (a) strengthening the stem
 - (b) manufacturing food
 - (c) conducting manufactured food
 - (d) none of the above.
2. People suffering from myopia
 - (a) can see near objects clearly
 - (b) can see far away objects clearly
 - (c) cannot see any object clearly
 - (d) are colour-blind
3. The cilia in paramecium are used for
 - (a) respiration
 - (b) locomotion
 - (c) protection
 - (d) excretion
4. Which of these types of skeleton is most appropriate to the cockroach?
 - (a) hydrostatic skeleton
 - (b) exoskeleton
 - (c) endoskeleton
 - (d) cartilaginous skeleton
5. When proteins are broken down they provide
 - (a) oxygen
 - (b) carbohydrate
 - (c) energy
 - (d) amino acids

6. The function of lenticel is
(a) to receive excess water in the plant
(b) to absorb water from the atmosphere
(c) for gaseous exchange
(d) to absorb light
7. Which of the following is characteristic of the animal cell
(a) presence of chloroplasts
(b) possession of a cellulose cell wall
(c) absence of large vacuoles
(d) presence of large vacuoles
8. In the life history of *Schistosoma* (Bilharzia), of the following is the intermediate host
(a) man
(b) snail
(c) mosquito larva
(d) fish
9. The hormone which tones up the muscles of a person in the time of danger is from the
(a) thyroid gland
(b) pancreas
(c) adrenal gland
(d) spleen
10. The study of the organisms and the environment of an abandoned farmland is the ecology of
(a) a community
(b) a population
(c) a species
(d) an ecosystem
11. At fertilization
(a) one chromosome from the male joins another from the female
(b) one gene from the male combines with the other from the female
(c) the male nucleus fuses with the female nucleus
(d) one set of chromosomes combines with another set from the female
12. The neck region of the tapeworm (*Taenia* spp.) is responsible for the

- (a) production of eggs
- (b) the storage of eggs
- (c) the formation of new segments
- (d) the development of the suckers.

13. The movement of molecules from a region of higher concentration to one of lower concentration is

- (a) diffusion
- (b) transpiration
- (c) osmosis
- (d) plasmolysis

14. The region of cell division in a root is

- (a) root cap
- (b) endodermis
- (c) xylem
- (d) Meristem

15. Which of the following is not an excretory organ?

- (a) Luns
- (b) kidney
- (c) leaf
- (d) large intestine

16. The part of the mammalian brain responsible for maintaining balance is

- (a) medulla oblongata
- (b) cerebellum
- (c) optic lobe
- (d) cerebrum

17. A sugar solution was boiled with Fehling's solutions A and B and the colour remain blue. The sugar tested was

- (a) glucose
- (b) maltose
- (c) fructose
- (d) sucrose

18. The blood vessel which carries digested food from the small intestine to the liver is

- (a) renal vein
- (b) renal artery
- (c) hepatic artery
- (d) hepatic portal vein

19. The maize grain is regarded as a fruit and not a seed because

- (a) it is covered by a sheath of leaves
- (b) the testa and fruit wall fuse after fertilization
- (c) it has both endosperm and cotyledon
- (d) the pericarp and seed coat are separate

20. Identical twins are produced under one of the following conditions

- a. Two ova fertilized at the same time by the sperm
- b. One ovum fertilized, divides to give two embryos
- c. Two ova fertilized by one sperm
- d. One ovum fertilized by two sperms

21. Which of the following hormones is produced during fright or when agitated?

- (a) insulin
- (b) adrenalin
- (c) thyroxine
- (d) pituitary

22. Grasses — Grasshopper — Lizards — Snakes

— Hawks. In the food chain, the organisms which are the least in number are

- (a) grasses
- (b) hawks
- (c) lizards
- (d) snakes

23. One significant difference between roots and stem is that

- a. branch root originate in the pericycle while branch stems do not.
- b. Stems are always below the ground while roots are always under the ground
- c. Stems are positively geotropic while roots are negatively geotropic
- d. Stems are sometimes used for storage while roots are never so used.

24. The arrangements below are steps in protein digestion. Which is the correct sequence? A — polypeptides, B — protein, C — amino acids, D — peptones

- (a) a—>b---c—>d
- (b) c—d---a—b
- (c) b—>c—>a—>d
- (d) b—>d—>a—c

25. Partially digested food ready to leave the stomach is called

- (a) chime
- (b) curd
- (c) glycogen
- (d) paste

BIOLOG 2012/2013 ANSWER

1.C	6.C	11.B	16.C	21.B
2.A	7.C	12.C	17.A	22.B
3.B	8.B	13.B	18.D	23.A
4.B	9.C	14.A	19.B	24.D
5.D	10.A	15.D	20.B	23.A

QUOTABLE QUOTES

- Successful people are very clear about **who they are**, **where they are going** and **what they want**.
- Set high goals and standards for yourself; resist temptation of the comfort zone.
- Your brain has more than **100billion cells**. It has the capacity to take you from **failure to success** and/or from **rag to riches**.

We wish all Nigerian students goodluck in their pursuit of academic excellence.

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